The Unique Games Theorem

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Abstract

We prove the Unique Games Conjecture. For every fixed $\varepsilon, \delta \in (0, 1/2)$, we give a deterministic polynomial-time reduction from 3SAT to Unique Games over a fixed finite alphabet, with completeness at least $1 - \varepsilon$ and soundness at most δ. The construction uses a new noise law for the degree-two matrix shortcode. An equivariant nonlinear evaluation is almost invariant under this noise, whereas every family of linear observations whose restrictions to the answer alphabet have sufficiently high rank detects the noise with constant probability. Fourier comparison and advice decoding combine this separation with known Grassmann expansion and projection-game repetition.

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1 Introduction

A permutation constraint prescribes exactly one label at one endpoint for each label at the other. The Unique Games Conjecture asserts that systems of such constraints can be hard to approximate even when almost all constraints can be satisfied. We resolve this conjecture positively, with an explicit unweighted reduction from 3SAT.

1.1 The theorem

A *Unique Games instance* has a finite vertex set, a finite alphabet Σ, and a nonempty list of oriented constraints $e = (u_e, v_e, \pi_e)$, where π_e is a permutation of Σ. A labeling a satisfies e when $a(v_e) = \pi_e(a(u_e))$. Its value is the fraction of satisfied constraints, and $\mathrm{val}(G)$ is the maximum value of a labeling. The instance is *explicit and unweighted* when the entire vertex and constraint lists are written out, including the table of each permutation. A simple bipartite instance has two disjoint vertex sets and at most one constraint on each pair of opposite vertices.

Theorem 1.1 (Unique Games Theorem)**.** *For every fixed $\varepsilon, \delta \in (0, 1/2)$, there are an integer $s \geq 1$ and a deterministic polynomial-time reduction that maps each* 3SAT *formula φ to an explicit unweighted Unique Games instance G_φ over $\Sigma = \mathbb{F}_2^s$, with*

$$\begin{aligned} \varphi \text{ satisfiable} &\implies \mathrm{val}(G_\varphi) \geq 1 - \varepsilon, \\ \varphi \text{ unsatisfiable} &\implies \mathrm{val}(G_\varphi) \leq \delta. \end{aligned}$$

The graph is simple and bipartite, and every constraint is a translation $a(v_e) = a(u_e) + c_e$ of Σ. The alphabet, the degree of the running-time polynomial, and its constants depend only on ε, δ.

Thus the result resolves the Unique Games Conjecture positively in its usual edge-value formulation [23, Conjecture 2.5]. The two errors are prescribed independently, before the alphabet is chosen. Binary translation constraints are a known equivalent formulation of the conjecture [24, Corollary 13 and Section 11.1]; the reduction here produces that form directly.

Theorem 1.1 supplies the Unique Games hypothesis in earlier hardness reductions for cut, covering, constraint-satisfaction, ordering, deletion, and clustering problems [24, 27, 36, 17, 9, 10]. The reductions and the resulting approximation thresholds are due to those authors. Section 9 gives the precise problem formulations, strict thresholds, and reduction qualifications, together with proofs of the consequences.

1.2 Proofs, codes, and the quantifiers

The central difficulty is to retain completeness near one when every label permits only one reply. The earlier connection between proof verification and approximation hardness was developed by Feige, Goldwasser, Lovász, Safra, and Szegedy [15], by Arora and Safra through proof composition [3], and by Arora, Lund, Motwani, Sudan, and Szegedy through the constant-query PCP theorem [2]. Bellare, Goldreich, and Sudan introduced the long code and folding for verifier composition [7, Section 3.3]. The long code of an answer a is the table $f \mapsto f(a)$, indexed by Boolean functions on the answer set. In Håstad's parity-hardness proof, Fourier analysis recovers an outer-game strategy from arbitrary proof tables that make his verifier accept sufficiently often, without first recovering nearby codewords. His nearly satisfiable parity gap is our starting hardness input [18].

Khot introduced the Unique Games Conjecture in 2002 and gave an algorithm based on semidefinite rounding [22]. Several algorithmic guarantees explain why its order of quantifiers matters. For a fixed target error, Trevisan combines semidefinite rounding with graph decomposition

to obtain a polynomial-time guarantee when the input error shrinks at the reciprocal-logarithmic rate in the instance size [41]. Charikar, Makarychev, and Makarychev use randomized semidefinite rounding to obtain expected value $1 - O(\sqrt{\epsilon \log q})$ from a q-label instance of value at least $1 - \epsilon$ [8]; this guarantee loses accuracy as the alphabet grows, whereas the conjecture permits a large fixed alphabet after the errors are chosen. Kolla's spectral enumeration gives a running-time bound in terms of the dimension of a suitable spectral subspace of the label-extended graph, whose vertices are variable–label pairs [30]. Arora, Barak, and Steurer use spectral decomposition and enumeration to obtain near-satisfying solutions for sufficiently small fixed completeness error, with a guarantee independent of the fixed alphabet size [1]; their running-time bound is subexponential, not polynomial, for fixed positive error.

Integrality gaps concern the strength of a relaxation. Khot and Vishnoi constructed Unique Games with semidefinite value arbitrarily close to one and integral value arbitrarily close to zero, even with triangle inequalities [29]. Barak, Gopalan, Håstad, Meka, Raghavendra, and Steurer introduced the short code, based on Reed–Muller codes, as a shorter substitute for the long code in applications including alphabet reduction and quantitative integrality gaps [5]. These constructions constrain particular relaxations; a hardness reduction requires a separate argument. Bafna, Barak, Kothari, Schramm, and Steurer obtain polynomial-time constant-value guarantees for nearly satisfiable translation games whose graphs have suitable low-degree sum-of-squares hypercontractivity certificates, including noisy hypercube and noisy Reed–Muller short-code graphs [4]. The rank-one matrix shortcode used below is a different graph.

1.3 Grassmann expansion and the completeness obstacle

The closest preceding hardness breakthrough concerns 2-to-1 and 2-to-2 games with imperfect completeness. A 2-to-1 constraint is a projection with exactly two preimages per right label. A 2-to-2 constraint partitions the labels on each side into pairs and matches these pairs bijectively, allowing the four combinations within a matched pair. Khot, Minzer, and Safra's STOC 2017 work proposed a Grassmann-graph reduction whose nonstandard soundness guarantee was conditional on a structural hypothesis [26]. Dinur, Khot, Kindler, Minzer, and Safra's STOC 2018 work proposed a reduction with ordinary edge-satisfaction soundness, conditional on an agreement hypothesis [12]; their expansion work formulated the relevant structural problem and established partial cases [11]. Barak, Kothari, and Steurer connected expansion to the needed agreement test through a binary matrix representation [6]. The expansion theorem of Khot, Minzer, and Safra completed this line in 2018 [25].

These results established hardness for almost satisfiable 2-to-1 and 2-to-2 games and, as a consequence, a Unique Games gap with completeness close to one half and arbitrarily small soundness. Obtaining Unique Games completeness arbitrarily close to one was a separate problem.[1]

The degree-two matrix shortcode of Barak, Kothari, and Steurer [6] is the formulation used here: its vertices are binary matrices and its steps add rank-one matrices. Their two-reply rule permits either the old answer or its translate by the update's left factor; requiring equality alone gives a one-half completeness guarantee. They discussed higher-degree tests as a possible route to improving this completeness [6, Section 1.6]. The inverse statement we use is the soundness of their equality test [6, Hypothesis 2.1], which follows from Grassmann expansion; it appears in Theorem 4.1. Ellis, Kindler, and Lifshitz later gave a shorter expansion proof using hypercontractivity on spaces of linear maps [14]; our external input remains the Khot–Minzer–Safra theorem. Our proof keeps the

[1] A separate companion proves hardness for exact 2-to-1 games with perfect completeness and arbitrarily small fixed soundness, over fixed alphabets [34, Theorem 1.1].

established degree-two inverse theorem and changes the noise using a nonlinear map. The following comparison explains that choice.

1.4 The construction and proof

The nonlinear evaluation. For the uniform rank-one test, a nonzero evaluation $M \mapsto Mz$ changes by $a(l^\top z)$ under $M \mapsto M + al^\top$. Here M is a binary ℓ-by-m matrix, $z \in \mathbb{F}_2^m \setminus \{0\}$, and $a \in \mathbb{F}_2^\ell$, $l \in \mathbb{F}_2^m$ are independent and uniform. This evaluation passes the equality test with probability $(1 + 2^{-\ell})/2$. We replace this evaluation by a nonlinear one that is almost unchanged by a different noise law. The noise must still be detected by sufficiently many independent linear observations: this is what will let soundness return to the ordinary shortcode test.

For any prescribed $p_* > 0$, Lemma 3.3 fixes a rank threshold r_* and, for every sufficiently large ℓ, constructs a binary space $\mathcal{V}$ containing $K = \mathbb{F}_2^\ell$, a map $C : \mathcal{V} \to K$, and a noise law μ on $\mathcal{V}$ such that

$$C(x + h) = C(x) + h \quad (x \in \mathcal{V},\ h \in K), \qquad \Pr[C(x + a) \neq C(x)] \leq p_*,$$

where in the probability x is uniform and independent of $a \sim \mu$. Every family of linear functionals whose restrictions to K have rank at least r_* detects a with probability at least $1/8$. The threshold r_* depends only on p_*, leaving ℓ free to meet the later shortcode and folding requirements. For uniform $P \in \mathrm{Hom}(\mathbb{F}_2^m, \mathcal{V})$, the new evaluation $C(Pz)$ changes to $C(Pz + a(l^\top z))$ under the rank-one update. Since Pz is uniform and $l^\top z$ is an independent fair bit, its equality test fails with probability at most $p_*/2$.

The separation between nonlinear stability and linear detection is proved in Section 6. A quadratic block over a finite field gives a nonzero output perturbation a fixed probability of disappearing. Concatenating these blocks, with noise on one random leaf, makes the nonlinear change probability decay geometrically with the depth (Lemma 6.6). For linear detection, we track the rank of a subfamily of observations down the same branch. Random changes of coordinates make the surviving space of linear functionals likely to satisfy the following bound: at rank r, restriction loses at most one dimension, with loss probability at most a constant times r times the block's cancellation probability. Taking the field large controls the exceptional spaces. A one-dimensional loss reduces the sum $\sum_{j=1}^{r} 1/j$ by $1/r$, cancelling that rank factor. Starting with sufficiently large rank therefore permits enough levels to make the nonlinear change small while leaving positive rank with constant probability. A surviving linear functional detects the uniform leaf noise with constant probability (Proposition 6.8). Finally, copies indexed by injections of the smaller output space into K allow ℓ to grow with r_* fixed: a high-rank family restricts onto the whole smaller dual for most injections (Section 6.5).

The permutation test and its completeness. Section 3 uses this evaluation to encode tuples $U = (e_1, \ldots, e_k)$ of binary parity equations on three distinct variables. A table is an affine map from the product of their solution planes to $\mathcal{V}$, represented by a matrix with $m = 1 + 2k$ columns, including the intercept. We adapt Khot–Safra's virtual tables [28, Section 1.1] by identifying tables that induce the same affine map on the same retained coordinates. Thus a projected table and its pullback have the same answer. Folding by constant K-translations, following Bellare–Goldreich–Sudan [7, Section 3.3], assigns one label to each translation orbit. Checking equality of the restored answers at P and $P + al^\top$ is then a translation constraint on those labels. The equivariance of C makes nonlinear evaluation compatible with these labels. A global parity assignment of error ξ supplies one common evaluation vector on every tuple it satisfies, so the matrix game has value at least $1 - k\xi - p_*/2$ (Lemma 3.4).

Soundness: reduction to the shortcode test. For soundness, suppose a labeling passes the matrix test with probability at least 0.99. A matrix frequency consists of one linear functional per column. Averaging over l makes its noise eigenvalue the probability that all these functionals vanish on a. The detection bound makes this nonnegative eigenvalue at most $7/8$ when their restrictions to K have rank at least r_*. High acceptance therefore forces a fixed amount of Fourier mass below that rank. Under uniform noise in K, the eigenvalue at restriction rank r is 2^{-r}, so this mass gives positive acceptance of the ordinary rank-one test (Lemma 7.1). Split $\mathcal{V} = K \oplus W$ and write $P = (M, T)$ for its two components. The ordinary test fixes T; on a positive fraction of pairs (U, T), Theorem 4.1 therefore gives agreement with an affine evaluation $Mz + u$, for some $z \in \mathbb{F}_2^m$ and $u \in K$, on a nonempty slice defined by at most r_{s} row equations and r_{s} column equations, for a fixed r_{s}.

Row advice and agreement on slices. To turn these slices into outer-game strategies, we need a positive fraction of advice admitting agreement on a slice. The equation prover's advice is (A, T, AM), where $A : K \to \mathbb{F}_2^{r_{\mathrm{s}}}$ is independently uniform; it specifies the row fiber $\{M' : AM' = AM\}$. Lemma 7.2 shows that a fixed positive fraction of such advice admits a slice with substantial agreement. Its erasure argument is the key: if the union of row fibers admitting such agreement were too small, random replacements of their entries could be chosen to preserve enough shortcode acceptance while destroying the agreement promised by the inverse theorem on every slice. The inverse theorem applies to the modified function and gives a contradiction. Each such advice then determines a fixed choice of slice and target. Writing its column equations as $Mq_j = t_j$, we can vary z by $\operatorname{span}\{q_j\}$ while absorbing the change into u. Folding forces the resulting coset to contain valid answers to U; these give a nonempty list of at most $2^{r_{\mathrm{s}}}$ answers.

Transfer to sparsely projected questions. We compare U with a question O obtained by independently replacing each equation by one uniformly chosen variable with probability β. This uses Khot–Safra's smooth outer-game method [28, Section 3 and Lemma 3.1], with correlated advice as in Khot–Minzer–Safra [26, Sections 3.3–3.4], also used by Minzer–Zheng [31, Section 3]. The referee samples a uniform table on O, gives its W-component and random linear combinations of its K-valued rows as advice there, and gives their pullbacks as advice on U, with the same map A. With all spaces fixed, take k large and $\beta = k^{-2/3}$. Since $k\beta^2 \to 0$, the full pulled-back table law, averaged over projection choices, approaches the unrestricted uniform law (Lemma 7.3). A chosen slice can become empty on projection; comparing the full table laws transfers both membership in that slice and agreement with its target. Consequently the chosen slice survives and retains agreement with positive probability (Lemma 7.4).

Decoding. On advice admitting such a slice, the U-strategy samples uniformly from its list. Adapting Khot–Safra's Fourier-list decoding [28, Section 1.1], the O-strategy samples Fourier mass on its own row fiber. Agreement on the transferred slice puts positive mass on frequencies that decode into the projected list. Each strategy uses only its own question and advice; the transferred slice certifies their success in the analysis. The independent sample from the bounded list then gives full answer agreement with probability at least a constant $\gamma > 0$, independent of k (Lemma 7.5 and Proposition 4.3).

The bound from vanishing advice slopes. On a NO parity instance, the advice game has vanishing value. Section 8 restricts to singleton coordinates where all advice slopes are zero. Each singleton has a fixed positive probability that its advice slopes vanish, and the same sparse

choice of β leaves $k\beta = k^{1/3} \to \infty$. These coordinates contribute neither slopes nor intercept corrections to the advice. Fixing A, the positions of these coordinates, the original intercept, and the other coordinate data therefore leaves independent equation–variable questions. Each prover can reconstruct its original input from its own questions on these coordinates and the fixed data, so parallel repetition applies to these coordinates. Averaging over their number bounds full advice agreement by $\exp(-ck^{1/3})$, for a constant $c > 0$ depending only on the fixed advice spaces (Lemma 4.4). This contradicts the decoded lower bound for large k, giving the fixed matrix-game gap in Proposition 4.5.

Completing the reduction. Section 5 rounds the weights and replaces each constraint occurrence by a path of four edges, producing a simple bipartite game while preserving a fixed gap and the translation form. The final step uses the same projection-game repetition bound as the argument using vanishing advice slopes, namely Dinur–Steurer's alphabet-independent bound [13, Corollary 1.2].[2] Its exponent t can be chosen from the fixed gap and δ before the answer alphabet K; the t-fold repeated game then has the alphabet $\Sigma = K^t$ of Theorem 1.1. We then choose the nonlinear change bound and alphabet, the tuple length k, and finally the parity error, making the completeness losses small enough for the prescribed ε. All dimensions and repetition counts are fixed by the two errors, so enumerating the finite experiments gives the deterministic polynomial-time reduction of Theorem 1.1.

Figure 1 displays the reduction and the two incompatible bounds that establish soundness.

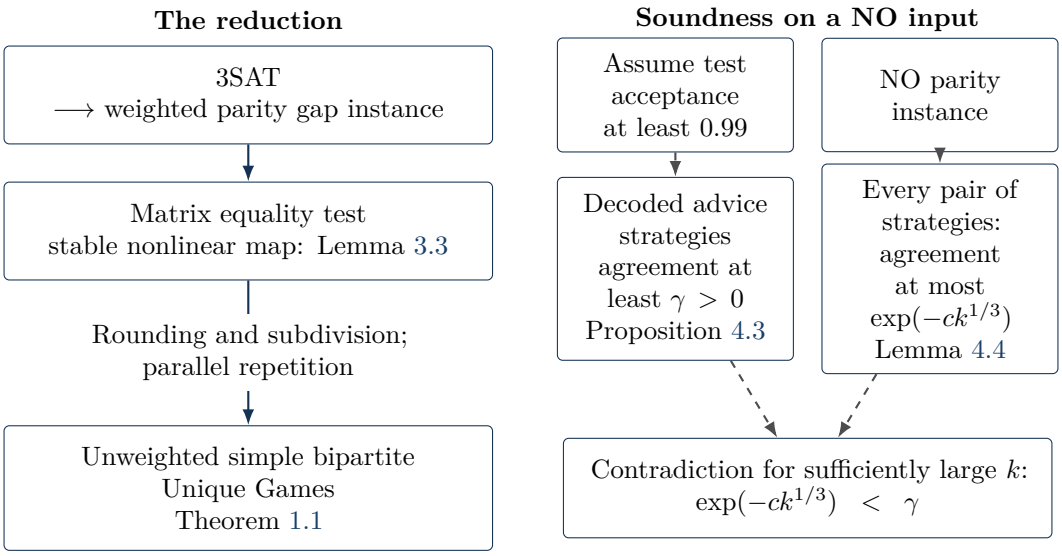


Figure 1: The solid arrows describe the reduction; the dashed arrows describe its soundness argument. The decoding lower bound follows from high test acceptance, while the upper bound from vanishing advice slopes follows from the NO parity gap and applies to every advice strategy. The constants $c, \gamma > 0$ are fixed before the tuple length k grows.

Organization. Section 2 fixes the conventions and the repetition bound. Section 3 states the properties of the equivariant map with additive noise and constructs the permutation test. Section 4

[2] Raz proved exponential decay for general two-prover games [39], and Holenstein simplified the proof [20]. Those bounds depend on the answer alphabet; Rao removed that dependence for projection games [38, Theorem 4].

states the two bounds on advice agreement and deduces the matrix-game gap; Section 5 then fixes the parameters and proves Theorem 1.1 from these inputs. The three supporting branches are proved in Sections 6, 7, and 8: the construction of the equivariant map with additive noise, matrix decoding, and the bound from vanishing advice slopes, respectively. Section 9 gives the approximation consequences. Appendix A proves the simple formulation with completeness on left vertices and the finite rational conversions used there.

2 Conventions

Weighted games. Intermediate instances may have loops, repeated constraint occurrences, and positive rational weights summing to one. Their value is the maximum total weight satisfied by a labeling. Occurrences retain separate IDs even if their constraints coincide. All rational numbers are encoded by binary numerators and denominators.

A bipartite *projection game* has a question for each vertex on either side, finite answer sets, and a map from left answers to right answers on each constraint. A strategy chooses one answer to each question. Answer sets that depend on the question can be identified with fixed alphabets of the same size; we use this for the four satisfying triples of a parity equation. Each prover uses only its own question. Private or shared randomness independent of the questions gives a mixture of deterministic strategies and cannot increase value.

The parallel power $G^{\otimes t}$ samples t independent constraint occurrences, sends the ordered question tuples, and accepts if every coordinate accepts. Answers may depend on the prover's entire tuple. Repeated question IDs retain separate coordinate positions and separate local answer copies.

Binary linear algebra and Fourier analysis. Except where a larger field is specified, all vector spaces and maps are finite-dimensional over $\mathbb{F}_2$. Write $\mathrm{Hom}(E, F)$ for the space of linear maps and $E^* = \mathrm{Hom}(E, \mathbb{F}_2)$ for the dual. A subspace $S \le E^*$ restricts to its image $S|_K \le K^*$ on a subspace $K \le E$; its restriction rank is $\dim(S|_K)$.

For a real function f on a binary space E, use normalized Fourier coefficients

$$\widehat{f}(g) = \mathbb{E}_{x \in E} f(x)(-1)^{g(x)}, \qquad f(x) = \sum_{g \in E^*} \widehat{f}(g)(-1)^{g(x)}, \qquad \sum_g |\widehat{f}(g)|^2 = \mathbb{E}_x |f(x)|^2.$$

These identities follow because a nonzero linear functional has equally many zero and one values, so its character has mean zero. The characters are therefore an orthonormal basis. On a space of matrices, this is the entrywise pairing with its dual matrix space; we specify tensor notation when defining the Fourier decoder.

Probability and finite constructions. Uniform sampling from a vector space includes zero. Samples are independent unless a dependence is stated. For finite probability laws,

$$\mathrm{TV}(P, Q) = \frac{1}{2} \sum_x |P(x) - Q(x)|.$$

It bounds the difference of expectations of any $[0, 1]$-valued function. We condition only on events of positive probability.

Random experiments describe finite weighted constraint lists. The reduction enumerates those lists deterministically. Every auxiliary dimension and tuple length is fixed before the input formula varies; the final proof checks both this parameter order and the binary encoding cost. Since Theorem 1.1 fixes ε and δ, its proof needs only that these parameters and the finite data for the

equivariant map and noise law exist. We also show that they can be computed from rational errors by terminating searches: in the last assertion of Lemma 3.3, in the paragraph on effective constants after Theorem 4.1 (proved at the end of Section 7.1), in the paragraph on effective choices in the proof of Theorem 1.1, and in Section 7.6. The tuple length k is taken to be a cube only so that the parameter $\beta = k^{-2/3}$ of the projection experiment is rational in the search of Section 7.6. Fourier witnesses and decoders are used only to bound game value and need not be efficient.

2.1 Classical projection-game repetition

We use the following form of the parallel repetition theorem of Dinur and Steurer, for the classical value and the parallel power defined above.

Theorem 2.1 (Projection-game repetition)**.** *Let G be a finite bipartite projection game with an arbitrary probability distribution on its constraint occurrences, allowing parallel occurrences and arbitrary finite question and answer sets. For its independent parallel power $G^{\otimes n}$, every integer $n \geq 1$ and every $0 < g < 1$ satisfy*

$$\mathrm{val}(G) \leq 1 - g \quad \Longrightarrow \quad \mathrm{val}(G^{\otimes n}) \leq (1 - g^2/16)^n. \tag{2.1}$$

The bound is independent of the question-set and alphabet sizes.

This is Corollary 1.2 of [13]. Their Section 2.1 permits nonnegative edge weights and parallel edges, and Section 2.2 defines the independent product used here. Thus the theorem applies to any finite weighted question distribution in our constructions. Interchanging the provers, if necessary, matches the direction of the projection. Different answer sets can be embedded in one finite alphabet, rejecting answers outside the legal set for the question. Each constraint still has at most one accepting reply to a given answer, as required by their definition. Replacing an illegal answer by any legal answer cannot decrease success, in either the original game or its product, so this embedding preserves both values.

3 The outer questions and the permutation test

We construct the weighted permutation instance used in the reduction. Its vertices encode affine tables on tuples of parity equations. We identify two tables exactly when they induce the same affine map on the same retained coordinates, including the same intercept. This common representation will allow us to compare an equation tuple with a sparsely projected tuple in Section 7.

3.1 Weighted parity equations and distinct positions

A *weighted parity instance* consists of a finite variable set $\mathcal{X}$, a finite nonempty set $\mathcal{E}$ of equation occurrences, and positive rational weights ω_e summing to one. An occurrence is an ordered equation

$$x_{v_{e,1}} + x_{v_{e,2}} + x_{v_{e,3}} = b_e, \qquad b_e \in \mathbb{F}_2.$$

Its value is the weighted fraction of equations satisfied, and $\mathrm{OPT}(\mathcal{E})$ denotes the maximum value. Different occurrences have different IDs even if their ordered equations coincide.

Theorem 3.1 (Weighted parity gap)**.** *For every fixed rational $\xi > 0$, there is a deterministic polynomial-time reduction mapping Boolean satisfiability instances to weighted parity instances with binary-encoded rational weights and the gap*

$$\begin{aligned} &\text{YES:} \quad \mathrm{OPT}(\mathcal{E}) \geq 1 - \xi, \\ &\text{NO:} \quad \mathrm{OPT}(\mathcal{E}) \leq \tfrac{1}{2} + \xi. \end{aligned}$$

The running-time polynomial may depend on ξ.

This is the gap consequence of Håstad's construction in [18, Theorem 5.4 and its proof]. More specifically, the proof uses a fixed dyadic verifier error $\delta > 0$, gives completeness $1-\delta$ and soundness $(1+\delta)/2$, and enumerates verifier choices as equations with their rational probabilities as weights. Choosing $\delta \le \xi$ gives the displayed statement. The enumeration is a deterministic reduction; the verifier's random choices describe the resulting weighted instance.

Lemma 3.2 (Making positions distinct)**.** *From a weighted parity instance $\mathcal{E}$ one can deterministically construct, in polynomial time, an instance $\mathcal{E}'$ in which the three variable IDs of every occurrence are distinct. Its weights are positive rationals of polynomial bit length, and*

$$\mathrm{OPT}(\mathcal{E}') \ge \mathrm{OPT}(\mathcal{E}), \qquad \mathrm{OPT}(\mathcal{E}') \le \frac{1+\mathrm{OPT}(\mathcal{E})}{2} + \frac{3}{100}.$$

Proof. Replace each variable v by one hundred clones (v,c), $c \in [100]$. For each original occurrence, choose its three clones uniformly subject to choosing distinct clones whenever positions have the same underlying variable. List all the resulting occurrences and give them the corresponding conditional weights. There are at most 100^3 choices per original occurrence. Giving all clones their original variable's label proves the completeness assertion.

For the upper bound fix arbitrary clone labels, and let

$$t_v = \frac{1}{100}\sum_{c=1}^{100}(-1)^{x_{(v,c)}}.$$

First choose clones independently, without conditioning on distinctness. The satisfaction probability of occurrence e is

$$\frac{1+(-1)^{b_e}t_{v_{e,1}}t_{v_{e,2}}t_{v_{e,3}}}{2}.$$

Round each bias to a majority label, resolving zero biases arbitrarily. For every equation violated by this rounded assignment, the signed product in the numerator is nonpositive, so its satisfaction probability is at most $1/2$. The other equations have probability at most one. The weighted mean is therefore at most $(1+\mathrm{OPT}(\mathcal{E}))/2$. The probability of a forbidden collision among the independently chosen clones is at most $3/100$, by a union bound over the three pairs of positions. Conditioning on its complement changes the expectation of any $[0,1]$-valued function by at most $3/100$. This proves the bound. All conditional probabilities have constant-size denominators apart from their original occurrence weights. □

Applying Theorem 3.1 with $\xi < 1/100$ and then Lemma 3.2 gives completeness at least $1-\xi$ and NO optimum at most $4/5$.

For the rest of the construction, $\mathcal{E}$ denotes the preprocessed instance, so every equation has three distinct variable IDs. Its *ordinary equation-versus-variable game* samples e according to ω and j uniformly from $[3]$. The first prover receives the occurrence ID e and answers with a satisfying triple for that equation; the second receives the variable ID $v_{e,j}$ and answers with a bit. Acceptance means equality at position j. Because the IDs within an equation are distinct, the question pair determines the projection constraint.

For any second-prover strategy, its bits constitute a global assignment. On an equation violated by that assignment, every valid first-prover answer disagrees in at least one position. Hence this ordinary projection game, denoted $G_{\mathcal{E}}$, satisfies

$$\mathrm{val}(G_{\mathcal{E}}) \le 1 - \frac{1-\mathrm{OPT}(\mathcal{E})}{3} \le \frac{14}{15} \quad \text{if } \mathrm{OPT}(\mathcal{E}) \le \frac{4}{5}. \tag{3.1}$$

The answer alphabets have sizes four and two, respectively.

3.2 Tuples, local answer copies, and the coordinate for the intercept

Fix a positive integer k. An equation question $U = (e_1, \dots, e_k)$ is sampled from $\omega^{\otimes k}$, with replacement. Its answer space is the product of the k affine solution planes. Blocks always use local copies: even if the same equation or variable ID appears in several blocks, their local answers are separate coordinates.

For a parameter $0 < \beta < 1$, independently in every block either retain the equation, with probability $1 - \beta$, or retain just the variable ID at a uniformly chosen position, with probability β. The resulting question is denoted O, and π is the coordinate projection from answers on U to answers on O. The question O records an equation ID or a variable ID at every ordered block position. A singleton block accepts either bit.

Represent an equation answer by its first two bits (x_i, y_i); the third is $b_{e_i} + x_i + y_i$. A singleton is represented by its actual bit. Add a common coordinate t_0. Thus

$$L_U = \mathbb{F}_2^{1+2k}, \qquad m = 1 + 2k,$$

and L_O has the coordinate t_0, two coordinates for each retained equation, and one for each singleton. In either space, vectors whose first coordinate is one are exactly valid answers. Write $e_0^\top$ for the functional reading that coordinate.

The affine projection π extends to a linear surjection $\bar{\pi} : L_U \to L_O$ preserving t_0. On retained equation blocks it is the identity, and on a block projected to a singleton its coordinate is

$$\begin{cases} x_i, & \text{position 1,} \\ y_i, & \text{position 2,} \\ b_{e_i} t_0 + x_i + y_i, & \text{position 3.} \end{cases} \tag{3.2}$$

The coordinate projections obtained by these blockwise choices form a class of maps independent of β. Surjectivity follows by fixing t_0 and choosing coordinates independently in each block. We use the same conventions for questions with some blocks omitted.

3.3 An equivariant map with additive noise

The tables will take values in a binary space containing the answer alphabet. The following lemma supplies a nonlinear map that usually preserves its value under the test's noise, although sufficiently many independent linear observations detect that noise with constant probability.

Lemma 3.3 (An equivariant map with additive noise)**.** *For every $p_* > 0$ there is an integer r_* such that, for every sufficiently large integer ℓ, there are a finite-dimensional binary space $\mathcal{V}$ with a specified inclusion of $K = \mathbb{F}_2^\ell$, a map $C : \mathcal{V} \to K$, and a rational probability distribution μ on $\mathcal{V}$ satisfying*

$$C(x + h) = C(x) + h \qquad (x \in \mathcal{V},\ h \in K), \tag{3.3}$$

and, for independent uniform $x \in \mathcal{V}$ and $a \sim \mu$,

$$\Pr[C(x + a) \neq C(x)] \leq p_*. \tag{3.4}$$

Moreover, every linear subspace $G \leq \mathcal{V}^*$ with $\dim(G|_K) \geq r_*$ satisfies

$$\Pr_{a \sim \mu}[\text{there exists } g \in G \text{ with } g(a) \neq 0] \geq \frac{1}{8}. \tag{3.5}$$

For rational p_*, the threshold and all the finite data can be found by a terminating deterministic algorithm.

Section 6 constructs these data and proves the lemma. The equivariance identifies the given copy of K with output translations; it will make evaluation of a table compatible with folding.

3.4 Representing tables and folding

The identification of tables across projected questions follows the virtual-table approach of Khot–Safra [28, Section 1.1]. We first express each table on the coordinates through which it factors. This will give the same representation to a projected table and its pullback.

Fix the map and noise data $(\mathcal{V}, K, C, \mu)$ from Lemma 3.3, where $K = \mathbb{F}_2^\ell$, and fix a linear complement $\mathcal{V} = K \oplus W$. A *table* on a question Q is an affine map from its answer space to $\mathcal{V}$, or equivalently its unique linear extension $P \in \mathrm{Hom}(L_Q, \mathcal{V})$. Here Q may be an equation question or one of its coordinate projections, and the same definition applies to a question with omitted blocks. Write c for the coefficient of the coordinate t_0 in P.

In an equation block, inspect the rank of its slope map $\mathbb{F}_2^2 \to \mathcal{V}$, whose contribution we write as $a_i x_i + d_i y_i$. If its rank is two, retain the equation occurrence and this contribution. If its rank is zero, omit the block. In the remaining case the rank is one, so the coefficients have exactly one of the forms

$$(a_i, d_i) = (v, 0), \quad (0, v), \quad (v, v), \qquad v \neq 0.$$

These forms retain only the variable at position one, two, or three, respectively, with contribution vz_i in its actual bit z_i. For the third form,

$$v(x_i + y_i) = vz_i + b_{e_i} v t_0, \qquad z_i = b_{e_i} t_0 + x_i + y_i,$$

so this replacement also adds $b_{e_i} v$ to the intercept. In a singleton block, retain its variable ID and its original contribution exactly when its slope is nonzero; otherwise omit the block.

The retained blocks specify the *retained coordinates* of (Q, P). They keep their indices in $[k]$, their equation/variable tags, and their IDs. A retained equation keeps its ordered first two coordinates, and a retained singleton uses its actual-bit coordinate. Thus repeated IDs in different blocks still have separate local coordinates. The set of retained coordinates may be empty; the coordinate t_0 remains in every case.

Let J_3 be the set of equation blocks reduced to their third bit, and write v_i for the nonzero slope in such a block. The *induced map* $\widetilde{P}$ on the retained coordinates has the retained block contributions specified above and intercept

$$\widetilde{c} = c + \sum_{i \in J_3} b_{e_i} v_i. \tag{3.6}$$

To check the complete table, let L_{ret} be the space with the retained coordinates and t_0 and let $\rho_{Q,P} : L_Q \to L_{\mathrm{ret}}$ retain t_0 and the specified block coordinates. It is the identity on retained equations and on retained singleton blocks already present in Q, uses the corresponding actual-bit coordinate from Equation (3.2) on a reduced equation, and discards omitted blocks. Choosing the coordinates independently in each block shows that $\rho_{Q,P}$ is surjective. The block identities and Equation (3.6) give

$$P = \widetilde{P} \rho_{Q,P}.$$

Surjectivity makes $\widetilde{P}$ the unique linear map with this factorization. We define the *table representation* of (Q, P) to be its retained coordinates together with this entire induced map. Thus two tables have the same table representation exactly when their retained coordinates agree, with the same positions, tags, and IDs, and their induced affine maps agree on those coordinates, including their intercepts.

We next check compatibility with every coordinate projection $\bar{\pi} : L_U \to L_O$. Let $P' \in \mathrm{Hom}(L_O, \mathcal{V})$. An equation retained in O has the same contribution in P' and its pullback $P'\bar{\pi}$, so its reduction is identical. A singleton of slope zero disappears on both sides. A singleton of nonzero slope v pulls back to the corresponding one of the three rank-one patterns above, which reduces to

the same variable ID at the same block position. For a third-position singleton, the pullback adds $b_{e_i} v$ to the intercept, and its reduction adds $b_{e_i} v$ once more. The two terms cancel over $\mathbb{F}_2$. All other intercept corrections come from retained equation blocks and are identical on the two sides. The retained coordinates and complete induced maps therefore agree: $(U, P'\bar{\pi})$ and (O, P') have the same table representation.

Consider the set of representations of all tables on equation questions $U \in \mathcal{E}^k$. The compatibility just proved shows that the table representation of every table on a question obtained by a coordinate projection already belongs to this set, by pullback from any compatible U.

We now apply the folding principle of Bellare, Goldreich, and Sudan [7, Section 3.3]: values on one representative determine the values on its translates. Adding a constant $h \in K$ to a table representation means adding h to the intercept of its induced map, leaving its retained coordinates and slopes fixed. This is realized by replacing a presenting table P with $P + he_0^\top$, so it preserves the set of table representations. The action is free because the table representation records the intercept. A vertex of our permutation instance is an orbit of this action, represented by its unique table representation in which the induced map has intercept with zero K component.

For a table P on an equation question or a projected question Q, let $c(Q, P) \in K$ be the K component of the intercept in its table representation. Let $v(Q, P)$ denote the orbit vertex represented by the table representation obtained after removing this component. A labeling λ of the vertices defines the restored table answer

$$F_Q(P) = \lambda(v(Q, P)) + c(Q, P).$$

Adding a constant changes only the removed component, so

$$F_Q(P + he_0^\top) = F_Q(P) + h \qquad (h \in K). \tag{3.7}$$

Tables with the same table representation have the same representative and removed component. The pullback compatibility therefore gives, for every such $U, O, \bar{\pi}$ and every table P' on O,

$$F_U(P'\bar{\pi}) = F_O(P'). \tag{3.8}$$

3.5 The matrix equality test and completeness

The permutation instance, denoted $\mathcal{G}$, samples independently

$$U \sim \omega^{\otimes k}, \qquad P \text{ uniform in } \mathrm{Hom}(L_U, \mathcal{V}), \qquad a \sim \mu, \qquad l \text{ uniform in } \mathbb{F}_2^m,$$

and checks

$$F_U(P) = F_U(P + al^\top). \tag{3.9}$$

Here $al^\top$ denotes the map $z \mapsto a(l^\top z)$. Each sample specifies an edge between the orbits of table representations $v_0 = v(U, P)$ and $v_1 = v(U, P + al^\top)$. With the removed components $c_0 = c(U, P)$ and $c_1 = c(U, P + al^\top)$, the test reads $\lambda(v_0) + c_0 = \lambda(v_1) + c_1$, so its constraint is the permutation

$$\lambda(v_1) = \lambda(v_0) + c_0 + c_1$$

of the fixed alphabet K. Edge occurrences have their sampling probabilities as weights. At this stage loops and repeated edges are allowed. The sparse projection experiment involving O is used to analyze these same vertex labels.

Lemma 3.4 (Completeness of the outer matrix test)**.** *Let $\mathcal{G}$ be the matrix game just defined, with tuple length k and map and noise data $(\mathcal{V}, K, C, \mu)$. If $\mathcal{E}$ has an assignment of value at least $1-\xi$ and*

$$\Pr_{x \text{ uniform in } \mathcal{V},\ a\sim\mu}[C(x+a) \neq C(x)] \leq p_*,$$

then

$$\mathrm{val}(\mathcal{G}) \geq 1 - k\xi - \frac{p_*}{2}.$$

Proof. Fix a global bit assignment $\sigma : \mathcal{X} \to \mathbb{F}_2$ of value at least $1-\xi$. Give every set of retained coordinates a valid answer as follows. At a singleton use its global bit. At a retained equation use its first two global bits and set the third to the unique value satisfying the equation. These choices depend only on the retained coordinates, including their occurrence IDs, and therefore define an answer for every table representation independently of its origin.

At an orbit representative, evaluate its induced map on this answer and apply C. This defines a vertex labeling. The equivariance of C under K implies that its restored answer is evaluation of the actual induced map followed by C, in agreement with Equation (3.7).

With probability at least $1-k\xi$, every equation in U is satisfied by σ. On such a tuple, let $z_U \in L_U$ have $t_0 = 1$ and block coordinates $x_i = \sigma(v_{e_i,1})$, $y_i = \sigma(v_{e_i,2})$. These agree with the chosen answers on retained equations and on singletons at positions one or two. For a block reduced to its third bit, satisfaction gives

$$b_{e_i} + x_i + y_i = \sigma(v_{e_i,3}),$$

so the coordinate in Equation (3.2) is again the chosen global bit; this is where satisfaction of every equation of U is used. Thus every table's answer on the retained coordinates is $\rho_{U,P}(z_U)$, also when only the coordinate t_0 remains. Since $P = \widetilde{P}\rho_{U,P}$, the induced map takes the value Pz_U at this answer on the retained coordinates, so the restored answer is $F_U(P) = C(Pz_U)$. Consequently the two sides of Equation (3.9) are

$$C(Pz_U) \quad \text{and} \quad C(Pz_U + a(l^\top z_U)).$$

Since $z_U \neq 0$, the vector Pz_U is uniform in $\mathcal{V}$, independently of a, l, and $l^\top z_U$ is an independent fair bit. The test therefore fails with probability at most $p_*/2$ on these tuples. Adding the probability of a tuple containing an unsatisfied equation proves the bound. □

4 The matrix-game gap

The matrix game's soundness follows by comparing two bounds for the same advice experiment. Proposition 4.3 extracts local strategies with a fixed positive agreement probability from test acceptance at least 0.99. Lemma 4.4 makes every such agreement probability tend to zero on a NO parity instance. We state the inputs and give the deduction here; their proofs occupy Sections 7 and 8.

4.1 The shortcode soundness theorem

The structural input describes functions that often keep their value under a uniform rank-one perturbation. Such a function agrees with an affine evaluation on a set obtained by fixing a bounded number of linear combinations of rows and columns.

Write $\mathrm{Mat}_{\ell,m} = \mathrm{Hom}(\mathbb{F}_2^m, \mathbb{F}_2^\ell)$. A *row and column slice* is a nonempty set specified by equations

$$\mathcal{S} = \{M : d_i^\top M = s_i^\top \ (1 \leq i \leq n_r), \quad Mq_j = t_j \ (1 \leq j \leq n_c)\}, \tag{4.1}$$

where $d_i \in \mathbb{F}_2^\ell$, $s_i \in \mathbb{F}_2^m$, $q_j \in \mathbb{F}_2^m$, and $t_j \in \mathbb{F}_2^\ell$. The counts n_r, n_c need not be the ranks of their respective systems.

Theorem 4.1 (Equality soundness for degree-two shortcode)**.** *For every $\eta \in (0,1)$ there are $\alpha \in (0,1]$, an integer $r_{\mathrm{s}} \geq 1$, and an integer ℓ_0, depending only on η, with the following property. For every $\ell \geq \ell_0$ there is $m_0 = m_0(\eta, \ell)$ such that, for every $m \geq m_0$ and every function $f : \mathrm{Mat}_{\ell,m} \to \mathbb{F}_2^\ell$, if*

$$\Pr_{M,a,l}\big[f(M) = f(M + al^\top)\big] \geq \eta, \tag{4.2}$$

then there are a nonempty row and column slice $\mathcal{S}$ as in Equation (4.1) with $n_r, n_c \leq r_{\mathrm{s}}$, $z \in \mathbb{F}_2^m$, and $u \in \mathbb{F}_2^\ell$ such that

$$\Pr_{M|\mathcal{S}}\big[f(M) = Mz + u\big] \geq \alpha. \tag{4.3}$$

Here $M \in \mathrm{Mat}_{\ell,m}$, $a \in \mathbb{F}_2^\ell$, and $l \in \mathbb{F}_2^m$ are independent and uniform, and $M \mid \mathcal{S}$ is uniform on the slice. No folding condition is imposed on f.

Theorem 4.1 is the soundness statement of Barak, Kothari, and Steurer for the equality form of their degree-two shortcode test [6, Hypothesis 2.1 and Section 1.6], with the dimension requirements stated explicitly: first $\ell \geq \ell_0$, then $m \geq m_0(\eta, \ell)$. They derive it from their hypothesis on non-expanding sets of the shortcode graph by passing to a fiber of f [6, Hypothesis 1.8 and Lemma 2.3], and show that this hypothesis is equivalent to the Grassmann expansion hypothesis [6, Theorem 1.11], which Khot, Minzer, and Safra proved [25, Theorem 1.12]. The proof in Section 7.1 gives this derivation directly: it applies Grassmann expansion to a fiber of f in the matrix chart.

The constants and dimension thresholds in Theorem 4.1 can be chosen effectively, with rational α when η is rational. This follows from the quantitative bound in [25, Theorem 2.13 and Section 2.4] and the comparison estimates in [25, Lemmas 2.7–2.8], with theorem numbers referring to the 2018 revision. The proof in Section 7.1 gives explicit sufficient inequalities for these choices.

We now fix the parameters needed below. Choose $p_* > 0$ and the associated r_* from Lemma 3.3, and put $\eta = 2^{-r_*}/16$. For this η, take $\alpha, r_{\mathrm{s}}, \ell_0$ from Theorem 4.1. Then choose $\ell \geq \ell_0$ sufficiently large for Lemma 3.3 and for

$$2^{-(\ell - r_{\mathrm{s}})} < \alpha/8, \qquad 2^{-\ell} < \eta/4. \tag{4.4}$$

The value of η is the acceptance level at which Theorem 4.1 is applied: Lemma 7.1 gives average ordinary acceptance at least $0.9 \cdot 2^{-r_*} = 14.4\eta$, which leaves room for the erasure in Lemma 7.2. The first inequality in Equation (4.4) supplies slack for erasure and folding in Lemma 7.2 and for the Fourier estimate in Equation (7.11).

Obtain the map and noise data $(\mathcal{V}, K, C, \mu)$ at this value of ℓ, and fix a linear splitting $\mathcal{V} = K \oplus W$. All these objects are fixed before k grows. In particular, $\ell > r_{\mathrm{s}}$. We write a table as $P = (M, T)$, with $M : L_U \to K$ and $T : L_U \to W$.

4.2 The advice experiment and the decoding statement

We first specify the local inputs for two strategies whose answer agreement will contradict the outer-game gap. This experiment is defined for every positive integer k and every $0 < \beta < 1$. Let $\mathcal{Q}_k$ consist of all equation questions $U \in \mathcal{E}^k$ and all questions O obtained from them by the coordinate projections of Section 3. The questions U, O and the projection $\bar{\pi} : L_U \to L_O$ have the law in Section 3: independently in each block, an equation is retained with probability $1 - \beta$, and a uniformly chosen variable position is retained with probability β.

Definition 4.2 (Projected advice experiment)**.** Sample U, O, π, then an independent uniform map $A : K \to \mathbb{F}_2^{r_s}$ and independent uniform maps

$$T_O : L_O \to W, \qquad M_O : L_O \to K.$$

Put $S_O = AM_O$ and send the following inputs to the two provers:

$$\begin{aligned} &U\text{-prover:} \quad (U, A, T_O\bar{\pi}, S_O\bar{\pi}),\\ &O\text{-prover:} \quad (O, A, T_O, S_O). \end{aligned} \tag{4.5}$$

The map M_O beyond the stated advice is hidden. Each prover returns an actual answer, that is, a vector of first bit one in its coordinate space. They succeed when their answers a_U, a_O satisfy $\bar{\pi}a_U = a_O$.

The U-message contains the maps $(T_U, S_U) = (T_O\bar{\pi}, S_O\bar{\pi})$, obtained by padding maps sampled on O. We will first find advice admitting agreement on a slice under a different law: an unrestricted uniform table (M, T) on U, with $S_U = AM$. Later we compare the full padded table $(M_O\bar{\pi}, T_O\bar{\pi})$ with this unrestricted table after averaging over the projection choices. This comparison is the bridge from the extraction argument to the experiment.

Proposition 4.3 (Matrix decoding)**.** *Fix $p_* > 0$ and choose r_*, η, α, r_s, ℓ_0, ℓ, the map and noise data and the splitting $\mathcal{V} = K \oplus W$, in this order, as in the paragraph containing Equation (4.4); all of them are fixed before k. There are a constant $\gamma > 0$ and an integer $k_0 \geq 8$, depending only on these fixed parameters, with the following property.*

For every integer cube $k \geq k_0$ and every weighted parity instance with three distinct variable IDs per equation, suppose a function

$$F_Q : \mathrm{Hom}(L_Q, \mathcal{V}) \longrightarrow K$$

is given for every $Q \in \mathcal{Q}_k$. Assume folding for every such Q, every $P \in \mathrm{Hom}(L_Q, \mathcal{V})$, and every $c \in K$, and assume sharing for every coordinate projection $\bar{\pi} : L_U \to L_O$ and every $P' \in \mathrm{Hom}(L_O, \mathcal{V})$:

$$\begin{aligned} F_Q(P + ce_0^\top) &= F_Q(P) + c,\\ F_U(P'\bar{\pi}) &= F_O(P'). \end{aligned} \tag{4.6}$$

If this family passes the table test of Equation (3.9) with probability at least 0.99, then the projected advice experiment with $\beta = k^{-2/3}$ has local strategies whose probability of full answer agreement is at least γ.

Here, and in Lemma 4.4 and Proposition 4.5, k ranges over integer cubes so that $\beta = k^{-2/3}$ is rational; the search for k in Section 7.6 uses this to check its conditions exactly.

Every labeling of the vertices of $\mathcal{G}$ supplies such a family: its restored functions satisfy Equation (4.6) by Equations (3.7) and (3.8). The family is fixed when the two response rules are defined and may be built into those rules.

The proof of Proposition 4.3 is in Section 7. Its constant is uniform in the instance and in the table family, which allows the following upper bound to contradict it.

4.3 The upper bound and the gap

The upper bound uses the same fixed advice spaces as Definition 4.2; it applies to every pair of local strategies, independently of how they were obtained.

Lemma 4.4 (Agreement bound from vanishing advice slopes)**.** *Let $\mathcal{E}$ be a weighted parity instance with three distinct variable IDs in each equation and $\mathrm{OPT}(\mathcal{E}) \le 4/5$. For every integer cube $k > 1$, every pair of local randomized strategies in Definition 4.2, with $\beta = k^{-2/3}$, satisfies*

$$\Pr[\bar{\pi}(a_U) = a_O] \le \left(1 - \frac{\beta\, 2^{-\dim W - r_s}}{3600}\right)^k \le \exp\left(-\frac{2^{-\dim W - r_s}}{3600}\, k^{1/3}\right). \tag{4.7}$$

In particular, the upper bound tends to zero uniformly over the parity instance as k tends to infinity along cubes, with the advice spaces fixed.

Section 8 proves this bound by restricting to singleton coordinates on which the advice slopes vanish, and applying Theorem 2.1 to the remaining equation–variable questions.

4.4 Deduction of the matrix-game gap

We can now compare the upper bound for every advice strategy with the strategies extracted from a highly accepting matrix labeling.

Proposition 4.5 (A fixed gap with adjustable completeness)**.** *Fix $p_* > 0$ and choose r_*, η, α, r_s, ℓ_0, ℓ, the map and noise data and the splitting $\mathcal{V} = K \oplus W$, in this order, as in the paragraph containing Equation (4.4). There is a threshold, depending only on these fixed data, such that for every integer cube k above it and every weighted parity instance $\mathcal{E}$ with three distinct variable IDs per equation, the matrix game $\mathcal{G}$ of Equation (3.9) satisfies*

$$\begin{aligned} \mathrm{OPT}(\mathcal{E}) \ge 1 - \xi \quad &\Longrightarrow \quad \mathrm{val}(\mathcal{G}) \ge 1 - k\xi - p_*/2, \\ \mathrm{OPT}(\mathcal{E}) \le 4/5 \quad &\Longrightarrow \quad \mathrm{val}(\mathcal{G}) \le 99/100. \end{aligned}$$

The threshold is also independent of ξ.

Proof. Completeness is Lemma 3.4. For soundness, fix a labeling of $\mathcal{G}$. Its restored table functions satisfy the folding and sharing identities of Section 3. Proposition 4.3 supplies a constant $\gamma > 0$ and, for every sufficiently large cube k, gives local advice strategies with agreement at least γ whenever those functions have matrix-test acceptance at least 0.99. For all sufficiently large cubes k, the upper bound in Equation (4.7) is smaller than γ. Lemma 4.4 contradicts the existence of those strategies whenever $\mathrm{OPT}(\mathcal{E}) \le 4/5$. All thresholds are uniform over the parity instance. □

The order matters: p_* can be made arbitrarily small before the map and noise parameters are fixed, and ξ can be made arbitrarily small after k is fixed. Thus the same NO gap is compatible with any prescribed small completeness error.

5 Completion of the reduction

The matrix game has a fixed soundness gap and an arbitrarily small completeness error. We now obtain the form required by Theorem 1.1: unweighted, simple bipartite games with arbitrary prescribed errors. Rounding the occurrence weights and subdividing each occurrence will preserve a fixed gap. One application of alphabet-independent repetition then completes the proof.

5.1 Removing weights and repeated edges

We use the following elementary construction for weighted permutation constraints. At this intermediate stage, a constraint occurrence may be a loop, and several occurrences may join the same vertices.

Lemma 5.1 (Rounding and subdivision)**.** *Let H be a finite permutation-constraint game with $h \geq 1$ constraint occurrences and positive rational weights summing to one. For every rational $\tau > 0$, a deterministic construction produces an unweighted simple bipartite game B on the same alphabet, with $4\lceil h/\tau \rceil$ edges, such that*

$$\left| \mathrm{val}(B) - \left(1 - \frac{1 - \mathrm{val}(H)}{4}\right) \right| \leq \frac{\tau}{4}. \tag{5.1}$$

If the constraints of H are translations on a finite binary vector space, so are those of B. For fixed τ, when H is listed by its vertices, occurrences, permutation tables, and binary rational weights, the construction takes polynomial time in its encoding length.

Proof. Put $Q = \lceil h/\tau \rceil$ and write the occurrence weights as $w_1, \ldots, w_h$. Start with $n_e = \lfloor Q w_e \rfloor$, and add one to the $Q - \sum_e \lfloor Q w_e \rfloor$ entries having largest fractional parts, breaking ties in a fixed order. Then

$$n_e \geq 0, \qquad \sum_e n_e = Q, \qquad |w_e - n_e/Q| \leq 1/Q.$$

The weight differences sum to zero. Consequently, for every labeling, the change in the weight of its satisfied-occurrence set is at most

$$\frac{1}{2} \sum_{e=1}^{h} |w_e - n_e/Q| \leq \frac{h}{2Q} \leq \tau.$$

Replace each occurrence e by n_e copies, omitting it if $n_e = 0$. The resulting unweighted multigraph H' has Q occurrences and $|\,\mathrm{val}(H') - \mathrm{val}(H)| \leq \tau$.

Orient each occurrence e of H' from u to v, writing its constraint as $a_v = \rho_e(a_u)$. Replace it by the path

$$u \;-\!- p_e \;-\!- q_e \;-\!- r_e \;-\!- v,$$

using three fresh interior vertices for *every occurrence*. Place all original vertices and all q_e on the left, and all p_e, r_e on the right. In the direction from u to v, put identity permutations on the first three edges and ρ_e on the last. When expressing an edge in left-to-right orientation, invert its permutation if needed. Give all $4Q$ edges equal weight.

Fix labels at the original vertices. A satisfied occurrence extends to four satisfied edges. A violated occurrence cannot extend to four, but propagating labels along the first three edges satisfies exactly three. Since the interiors are disjoint, these extensions can be chosen independently. Maximizing over the original labels gives the exact identity

$$\mathrm{val}(B) = 1 - \frac{1 - \mathrm{val}(H')}{4}. \tag{5.2}$$

This proves Equation (5.1).

A loop becomes a four-cycle with three distinct new vertices. Parallel occurrences have disjoint interiors. Thus each left–right vertex pair carries at most one edge, so B is simple. An inverse binary translation is the same translation, and an identity is translation by zero; the claimed constraint form is preserved.

Finally, $Q \leq h/\tau + 1$ is polynomially bounded for fixed τ. Integer division and exact comparison of rational remainders compute the n_e in polynomial time and bit complexity. Listing their Q copies and the $3Q$ new vertices is therefore polynomial as well. □

5.2 The final reduction

Proof of Theorem 1.1. Fix $\varepsilon, \delta \in (0, 1/2)$. Choose positive rational $\varepsilon_0 \le \varepsilon$ and $\delta_0 \le \delta$. We will take $\tau \le 1/200$ when applying the rounding and subdivision lemma. For any matrix game $\mathcal{G}$ satisfying the NO bound $\mathrm{val}(\mathcal{G}) \le 99/100$, the resulting game B then satisfies

$$\mathrm{val}(B) \le 1 - \frac{1/100 - \tau}{4} \le 1 - \frac{1}{800}. \tag{5.3}$$

This gap does not depend on the alphabet. We can therefore use alphabet-independent repetition to choose its exponent before the construction of the equivariant map fixes that alphabet. Choose an integer $t \ge 1$ such that

$$\left(1 - \frac{1}{10240000}\right)^t \le \delta_0. \tag{5.4}$$

For example, $t \ge \lceil 10240000 \log(1/\delta_0) \rceil$ suffices. By Theorem 2.1, this exponent reduces every projection game of value at most $1 - 1/800$ to value at most δ_0.

Choose positive rational parameters

$$p_* \le \frac{\varepsilon_0}{t}, \qquad \tau \le \min\left\{\frac{1}{200}, \frac{\varepsilon_0}{t}\right\}. \tag{5.5}$$

Choose the rank threshold for linear detection r_* from Lemma 3.3, put $\eta = 2^{-r_*}/16$, and choose the corresponding shortcode parameters α, r_{s}, using a positive rational lower bound for α. Then choose a sufficiently large ℓ, as specified in Equation (4.4), and fix the corresponding map and noise data $(\mathcal{V}, K, C, \mu)$ and a splitting $\mathcal{V} = K \oplus W$, with $K = \mathbb{F}_2^\ell$.

Once the rational budgets are fixed, these choices are effective. For rational p_*, the effective assertion of Lemma 3.3 supplies all the finite map and noise data, including a computable rational noise law, by a terminating deterministic construction. The effective bounds stated after Theorem 4.1 supply the rational choice of α and the dimension thresholds, including $m_0(\eta, \ell)$ after ℓ has been fixed.

Next choose a cube $k \ge 8$ large enough for Proposition 4.5. Section 7.6 gives a terminating rational search for such a cube, checking the concentration, variation, and agreement bounds after the map, noise, and shortcode data are fixed.

Finally, with k fixed, choose a positive rational

$$\xi < \min\left\{\frac{1}{100}, \frac{\varepsilon_0}{2kt}\right\}. \tag{5.6}$$

Every choice depends only on the two prescribed errors. In particular, r_* is fixed before ℓ, the advice dimensions before k, and k before the parity error ξ.

Given a 3SAT formula, apply Theorem 3.1 and Lemma 3.2, and construct the matrix game $\mathcal{G}$ of Section 3. Proposition 4.5 gives

$$\begin{aligned} &\text{YES:} \quad \mathrm{val}(\mathcal{G}) \ge 1 - d, \quad d = k\xi + p_*/2 \le \varepsilon_0/t, \\ &\text{NO:} \quad \mathrm{val}(\mathcal{G}) \le 99/100. \end{aligned}$$

Apply Lemma 5.1 with τ to obtain the unweighted simple bipartite game B. It satisfies

$$\begin{aligned} &\text{YES:} \quad \mathrm{val}(B) \ge 1 - (d + \tau)/4, \\ &\text{NO:} \quad \mathrm{val}(B) \le 1 - 1/800, \end{aligned}$$

where the NO bound is Equation (5.3). The final output is $G_\varphi = B^{\otimes t}$. Its NO value is at most $\delta_0 \le \delta$, by Equations (2.1) and (5.4). On a YES input, use a labeling of B coordinatewise. A union bound gives

$$\mathrm{val}(G_\varphi) \ge 1 - \frac{t(d+\tau)}{4} \ge 1 - \varepsilon_0/2 \ge 1 - \varepsilon.$$

The output has the claimed graph and constraint form. Its two vertex sets are the t-fold products of the two sides of B. A pair of product vertices determines at most one edge in each coordinate, since B is simple, and hence at most one product edge. Independent uniform edge sampling is uniform on these product edges. The constraints of $\mathcal{G}$ are translations on K, by Section 3; rounding and subdivision preserve them. The product of translations with offsets $c_1, \dots, c_t$ is translation by $(c_1, \dots, c_t)$ on $K^t = \mathbb{F}_2^{\ell t}$. This is the alphabet $\Sigma = \mathbb{F}_2^s$ in the theorem, with $s = \ell t \ge 1$ fixed by the prescribed errors.

Polynomial encoding. Suppose the preprocessed parity instance has $N_{\mathcal{E}}$ positive equation occurrences. There are $N_{\mathcal{E}}^k$ equation tuples. The number of tables, noise outcomes, and perturbation vectors for each tuple is fixed after the parameter choices above, so the number h of listed matrix-test occurrences is a fixed constant times $N_{\mathcal{E}}^k$. Omit zero-probability outcomes and retain repeated occurrences separately. Each weight is a product of k parity weights, a fixed rational noise probability, and fixed uniform-sampling factors. It has polynomial binary length.

Table representations and folded representatives are computed by linear algebra in fixed dimensions, keeping the original variable and occurrence IDs. At most $2h$ endpoints appear. Even comparing every pair of table representations to identify equal vertices takes polynomial time; number the resulting distinct vertices consecutively. Largest-remainder rounding uses $Q = \lceil h/\tau \rceil = O(h)$ copies; subdivision creates $3Q$ new vertices and $4Q$ edges. The final product has $(4Q)^t$ edges and polynomially many vertices, because t is fixed. Full permutation tables on the fixed alphabet $\mathbb{F}_2^{\ell t}$ have constant size, and endpoint indices have $O(t \log(h+1))$ bits. These observations give a deterministic polynomial-time explicit construction throughout and complete the proof. □

6 An equivariant map with additive noise

We prove Lemma 3.3, whose equivariant map supplies the completeness labeling of the matrix game and whose linear detection bound supplies its spectral comparison. The construction separates a nonlinear observable from families of linear observables under the same additive noise. All vector spaces and duals in this section are over $\mathbb{F}_2$, unless a different field is indicated. Restriction of a subspace of a dual means its image under the restriction map.

We first construct a gadget with a fixed smaller output space, in which shifts induce output translations. After identifying shifts that induce the same translation, this gadget will detect every linear family whose restrictions fill the dual of the output space. We then combine copies indexed by injections of that space into K: a sufficiently high-rank family on K restricts onto the full output dual on most copies. This second step allows every sufficiently large dimension of K while keeping the rank threshold fixed.

For the recursive maps, Lemma 6.6 computes the nonlinear change probability, while Proposition 6.8 bounds detection by every linear family extending the pullbacks of all output functionals at the chosen height. Section 6.5 passes both conclusions through the quotient and then enlarges the output space; it contains the deduction of Lemma 3.3.

The fixed-output construction begins with a quadratic block. On an independent uniform input, a uniform nonzero perturbation in one of its update spaces can leave the nonlinear output unchanged.

For most subspaces of linear observations, restriction to the same update space loses at most one dimension, and the loss probability is bounded by three times the rank times this nonlinear stability probability. The sum $\sum_{j=1}^{r} 1/j$ in Equation (6.6) removes the rank factor from the expected loss per level. Random changes of coordinates make it unlikely that a descending family contains distinct nonzero elements proportional over the field or, for some binary linear map g, more than three times its dimension nonzero solutions z of $z \cdot g(z) = \sqrt{z_1 z_2 z_3}$, even when later linear maps depend on earlier choices. Concatenation therefore makes the nonlinear change probability small while leaving a constant probability of positive rank. At the final leaf, a surviving family detects an independent uniform symbol with constant probability.

6.1 The quadratic block

Fix an integer $r_0 \geq 1$, whose size will eventually depend only on p_*. Let $F = \mathbb{F}_{2^d}$, $q = 2^d$, and regard $X = B = F^3$ as binary spaces, each of dimension $b = 3d$. The integer d will be chosen sufficiently large in terms of r_0.

Use the trace pairing to identify both X^* and B^* with F^3: the vector $z \in F^3$ represents the linear functional $y \mapsto \mathrm{Tr}_{F/\mathbb{F}_2}(z \cdot y)$, where the dot denotes the field dot product. We abbreviate the trace by Tr. The trace pairing is nondegenerate, and $\mathrm{Tr}(t^2) = \mathrm{Tr}(t)$. Every element of F has a unique square root.

Define

$$Q(x_1, x_2, x_3) = (x_2 x_3, x_1 x_3, x_1 x_2), \qquad D(x, a) = Q(x + a) + Q(x) + Q(a). \tag{6.1}$$

The block observable is

$$C_{\mathrm{blk}} : X \oplus B \longrightarrow B, \qquad C_{\mathrm{blk}}(x, y) = y + Q(x).$$

It is equivariant under the vertical copy of B: $C_{\mathrm{blk}}(x, y + b') = C_{\mathrm{blk}}(x, y) + b'$ for $b' \in B$. For $z \in B^*$, the scalar observation $z \circ C_{\mathrm{blk}}$ changes by $z(b')$ under this shift. We shall compare it with linear functionals whose restriction to the vertical B is also z.

For a one-dimensional F-subspace $A = Fv$ of F^3, put

$$D_A = \{y \in B : v \cdot y = 0\}, \qquad U_A = \{(a, Q(a) + w) : a \in A,\ w \in D_A\} \subseteq X \oplus B. \tag{6.2}$$

These definitions are independent of the chosen nonzero generator v. There are $q^2 + q + 1$ such lines; set

$$\theta = \frac{1}{q^2 + q + 1}. \tag{6.3}$$

Lemma 6.1. *For each one-dimensional F-subspace $A \leq F^3$, the set U_A in Equation (6.2) is a binary subspace of dimension b, and the trace annihilator of D_A in $B^* = F^3$ is A. Moreover,*

$$D(x, a) \in D_A \qquad (x \in X,\ a \in A).$$

As A ranges over these lines, the subspaces U_A span $X \oplus B$.

Proof. The three coordinates of $D(x, a)$ are

$$x_2 a_3 + a_2 x_3, \qquad x_1 a_3 + a_1 x_3, \qquad x_1 a_2 + a_1 x_2.$$

When $a = tv$, every term in $v \cdot D(x, a)$ occurs twice, so their sum is zero. When both arguments are multiples of v, each coordinate of the polar expression is zero. Thus $Q|_A$ is binary additive, and the parametrization of U_A by $A \oplus D_A$ is a binary linear bijection. Its dimension is $d + 2d = b$.

Every $z \in A$ annihilates D_A under the trace pairing. The annihilator has binary dimension $3d - 2d = d$, so it is exactly A. Finally, for each coordinate line $A = Fe_i$, the space U_A contains $(Fe_i, 0)$, because $Q(te_i) = 0$, as well as $\{0\} \oplus D_A$. These three coordinate lines and their vertical hyperplanes span $X \oplus B$. □

The nonlinear response. The form of U_A makes the quadratic term of an update cancel. For $(a, Q(a) + w) \in U_A$,

$$C_{\mathrm{blk}}\big((x, y) + (a, Q(a) + w)\big) + C_{\mathrm{blk}}(x, y) = w + D(x, a).$$

The right-hand side lies in D_A by Lemma 6.1. The following calculation gives the exact law that the recursion will use.

Lemma 6.2 (Response to one block update)**.** *For every one-dimensional F-subspace $A \leq F^3$ and independent uniform $(x, y) \in X \oplus B$ and $\zeta \in U_A \setminus \{0\}$,*

$$\Pr\big[C_{\mathrm{blk}}((x, y) + \zeta) \neq C_{\mathrm{blk}}(x, y)\big] = 1 - \theta.$$

Proof. First take ζ uniform in all of U_A, still independently of the uniform input (x, y). Its parametrization $(a, Q(a) + w)$ has independent uniform $a \in A$ and $w \in D_A$. For every fixed x, a, the output difference $w + D(x, a)$ is uniform in D_A, so it is nonzero with probability $1 - q^{-2}$. The zero perturbation has mass q^{-3} and never changes the output. Conditioning on a nonzero perturbation therefore gives

$$\frac{1 - q^{-2}}{1 - q^{-3}} = \frac{q(q + 1)}{q^2 + q + 1} = 1 - \theta.$$

□

The lemma concerns an independent uniform input (x, y) and a uniform nonzero update. In the recursion constructed below, (x, y) will be the aggregate of the embedded child outputs and the update will come from a lower level; we will verify this conditional law before applying the lemma.

Linear restrictions. Fix a binary subspace $S \leq B^*$. A linear functional on $X \oplus B$ whose restriction to the vertical B is $z \in S$ has a unique form $(\gamma_z, z) \in X^* \oplus B^*$. Thus a binary-linear choice of one such extension for each $z \in S$ is specified by a binary linear map $\gamma : S \to X^*$. We ask which of these chosen extensions vanish when restricted to U_A. The restriction loses rank exactly when its kernel contains a nonzero element.

Lemma 6.3 (Kernel of a block restriction)**.** *Let $S \leq B^*$ and let $\gamma : S \to X^*$ be binary linear. For each line A, define*

$$\kappa_{A,\gamma} : S \longrightarrow U_A^*, \qquad \kappa_{A,\gamma}(z) = (\gamma_z, z)|_{U_A}.$$

Let $g : S \to F^3$ represent γ under the trace identification: $\gamma_z(x) = \mathrm{Tr}(g(z) \cdot x)$ for $z \in S$, $x \in X$. Then, for every nonzero $z \in S$,

$$\kappa_{A,\gamma}(z) = 0 \quad \Longleftrightarrow \quad A = Fz \quad \text{and} \quad z \cdot g(z) = \sqrt{z_1 z_2 z_3}.$$

Proof. If z lies in the kernel, it annihilates every vertical vector $(0, w)$ with $w \in D_A$. Lemma 6.1 therefore gives $z \in A$, and for $z \neq 0$ this means $A = Fz$. On the remaining vectors $(tz, Q(tz))$, $t \in F$, the functional has value

$$\mathrm{Tr}(t\, g(z) \cdot z + t^2 z_1 z_2 z_3). \tag{6.4}$$

The three summands of $z \cdot Q(tz)$ are equal, and three equals one in characteristic two. Also

$$\operatorname{Tr}(t^2 c) = \operatorname{Tr}(t\sqrt{c}).$$

Nondegeneracy of the trace pairing shows that Equation (6.4) vanishes for every t exactly when $z \cdot g(z) = \sqrt{z_1 z_2 z_3}$. Together with $z \in A$, this condition is sufficient as well: the vertical part is already annihilated. □

6.2 Two conditions on spaces of linear functionals

The kernel criterion suggests two sufficient controls. We shall use a family in which no field line contains two nonzero linear functionals and only a few linear functionals satisfy the equation $z \cdot g(z) = \sqrt{z_1 z_2 z_3}$ for any one choice of extensions. The uniformity over that choice will later allow a family of extensions to depend on earlier random changes of coordinates.

Definition 6.4. For $r \geq 1$, let $\mathcal{H}_r$ consist of the binary subspaces $S \leq F^3$ of dimension r such that no two distinct nonzero elements of S are proportional over F, and, for every binary linear map $g : S \to F^3$, at most $3r$ nonzero $z \in S$ satisfy

$$z \cdot g(z) = \sqrt{z_1 z_2 z_3}. \tag{6.5}$$

Write $H_0 = 0$ and $H_r = \sum_{j=1}^{r} 1/j$ for positive integers r. Fix one map $\gamma : S \to X^*$ before choosing a uniform line A, and let $r_A = \dim \operatorname{im} \kappa_{A,\gamma}$, where $r = \dim S > 0$. Lemma 6.3 now bounds the loss. If $S \in \mathcal{H}_r$, each field line contains at most one nonzero element of S, so the kernel has dimension at most one. For the fixed map g associated with γ, at most $3r$ linear functionals satisfy Equation (6.5); hence at most $3r$ lines can cause any rank loss. Consequently

$$r - r_A \leq 1, \qquad \Pr_A[r_A < r] \leq 3r\theta, \qquad \mathbb{E}_A[H_r - H_{r_A}] \leq 3\theta. \tag{6.6}$$

The last bound uses $H_r - H_{r-1} = 1/r$. For an arbitrary rank-r space, a loss still requires $A = Fz$ for some nonzero $z \in S$. Therefore

$$\Pr_A[r_A < r] \leq (2^r - 1)\theta \leq 2^r \theta, \qquad H_r - H_{r_A} \leq H_r. \tag{6.7}$$

These estimates concern rank under restriction. Detection of the eventual noise will follow only after a positive-rank family reaches a leaf.

It remains to show that subspaces in $\mathcal{H}_r$ are abundant.

Lemma 6.5. *Fix an integer $r \geq 1$. As $d \to \infty$, with $F = \mathbb{F}_{2^d}$, the probability that a uniform binary r-dimensional subspace of F^3 belongs to $\mathcal{H}_r$ tends to one.*

Proof. Choose independent uniform $x_{ij} \in F$, for $1 \leq i \leq 3$ and $1 \leq j \leq r$, and parametrize a binary linear map by

$$z_i(t) = \sum_{j=1}^{r} x_{ij} t_j, \qquad t \in \mathbb{F}_2^r.$$

Each nonzero input has uniform image in F^3, so the probability that this map is not injective is at most $(2^r - 1)q^{-3}$. Conditioned on injectivity, its image is a uniform binary r-subspace: every such subspace has the same number of ordered binary bases. Distinct nonzero binary inputs are linearly independent also over F, so their images are independent uniform elements of F^3. A union bound

therefore shows that the nonproportionality condition fails with probability $O_r(q^{-2})$, apart from the event of noninjectivity.

To prove the bound on the number of solutions simultaneously for all g, we eliminate the unknown basis values of g by a column-span condition. A nonzero minor will then bound its probability without taking a union over the field-dependent set of maps g. Fix a set E of more than $3r$ distinct nonzero inputs in $\mathbb{F}_2^r$. On an injective parametrization, the values of a binary linear g on a basis can be arbitrary vectors in F^3. Thus the equality on all of E implies that the vector

$$c = \big(\sqrt{z_1(t)z_2(t)z_3(t)}\big)_{t\in E}$$

lies in the F-column span of the matrix with $3r$ columns

$$M_{t,(i,j)} = t_j z_i(t), \qquad t \in E, \quad 1 \le i \le 3, \quad 1 \le j \le r.$$

Conversely, membership supplies the basis values of such a g. In particular, this column-membership condition already accounts for all possible maps g.

First regard the x_{ij} as independent indeterminates and set $L = \mathbb{F}_2(x_{ij})$. We claim that c is not in the column span of M even after extending scalars to an algebraic closure $\overline{L}$. The following r^3 elements are linearly independent over L:

$$\sqrt{x_{1j}x_{2k}x_{3l}}, \qquad 1 \le j,k,l \le r.$$

Indeed, after clearing denominators, a dependence would be a polynomial identity in the square roots of the indeterminates. The displayed monomials have distinct parity vectors, whereas coefficients polynomial in the x_{ij} have even exponents in those square roots. Terms from different parity vectors cannot cancel.

The expansion of c in these independent elements has coefficient vectors

$$v_{jkl} = (t_j t_k t_l)_{t\in E}.$$

Suppose that c belonged to the column span over $\overline{L}$. Every row vector over L annihilating M would then annihilate c. The independence just proved implies that it would annihilate every v_{jkl}. By double annihilation in the finite-dimensional space L^E, every v_{jkl} would belong to the column span over L itself.

For $n \ge 1$, let $V_n \le L^E$ be the evaluation span of all positive-degree monomials in the binary coordinates $t_1, \dots, t_r$ of degree at most n. Repeated indices in v_{jkl} give all monomials of degrees one and two as well as three, because $t_i^2 = t_i$ on the evaluation set. Each column $t \mapsto t_j z_i(t) = \sum_k x_{ik} t_j t_k$ of M is an L-combination of such monomials of degree at most two. The preceding conclusion and this form of the columns therefore imply

$$V_3 \subseteq \mathrm{col}_L(M) \subseteq V_2 \subseteq V_3.$$

These three spaces are equal. Multiplication by any t_i sends V_2 into V_3, so preserves their common value. Since this space contains the degree-one monomials, it contains evaluations of every positive-degree monomial. These evaluations span all of L^E: for any nonzero binary point, its usual indicator polynomial is a product of factors t_i and $1 + t_i$ with at least one factor t_i, and thus has no constant monomial. Its restriction to E gives the corresponding point indicator. Consequently the column span would have dimension $|E| > 3r$, a contradiction.

To pass from this algebraic statement to finite fields, square the entries of both M and c. The resulting matrices have polynomial entries, and the appended column has entries $z_1(t)z_2(t)z_3(t)$.

Let $h = \operatorname{rank}_L M \le 3r$. Squaring preserves rank over $\overline{L}$, so the augmented squared matrix has a nonzero minor of size $h+1$. This minor is a nonzero polynomial of degree at most $2h+3 \le 6r+3$. Indeed, the inputs $t \in E$ are fixed: a squared entry of M is quadratic in the indeterminates x_{ij}, and an appended entry is cubic. Each determinant term uses one appended entry and h other entries. The nonzero polynomial has coefficients in $\mathbb{F}_2$, so remains nonzero over every extension $\mathbb{F}_{2^d}$. All unaugmented minors of size $h+1$ vanish identically, so the unaugmented rank cannot increase on specialization. It follows that specialized column membership forces this particular nonzero polynomial to vanish. For uniform independent $x_{ij} \in F$, this event has probability at most $(6r+3)/q$ by the polynomial zero bound [40, Corollary 1]; see also the earlier sparse-polynomial formulation in [42, Section 3.1].

If the bound on the number of solutions fails, it fails on some set E of size $3r+1$. There are finitely many such sets depending only on r; if $2^r - 1 \le 3r$ there are none and the condition is automatic. A union bound, followed by conditioning on injectivity, proves the lemma. □

The defining conditions of $\mathcal{H}_r$ therefore control every possible map g. We now fix a field where subspaces outside $\mathcal{H}_r$ are rare at every rank $r \le r_0$. Combining the two bounds in Equation (6.7) bounds the expected decrease in H_r over a uniform line by $H_{r_0} 2^{r_0} \theta$ at ranks at most r_0. To limit the contribution from subspaces outside $\mathcal{H}_r$ of probability at most ε_0 to θ, choose a positive ε_0 with

$$H_{r_0} 2^{r_0} \varepsilon_0 \le 1, \qquad \varepsilon_0 < 1. \tag{6.8}$$

By Lemma 6.5, we may fix d sufficiently large that $b = 3d \ge r_0$ and, for every $1 \le r \le r_0$, a uniform rank-r subspace of B^* lies outside $\mathcal{H}_r$ with probability at most ε_0. In particular, there exists a subspace in $\mathcal{H}_{r_0}$, which we fix once and for all.

6.3 Concatenation and its noise

We keep the output space B fixed while concatenating the quadratic blocks. One noisy branch will be chosen at each level. The next lemma computes how fast the nonlinear change probability decays; we will then track the rank of linear restrictions along the same branch.

We construct binary input spaces P_t, shift subspaces $R_t \le P_t$, linear surjections $\lambda_t : R_t \to B$, and maps $C_t : P_t \to B$ such that

$$C_t(u+h) = C_t(u) + \lambda_t(h) \qquad (h \in R_t). \tag{6.9}$$

A shift $h \in R_t$ thus represents the output translation $\lambda_t(h)$; different shifts may represent the same translation. This form of the shift action is preserved by aggregation. At the end, quotienting by the shifts of value zero will turn the output translations into an actual subspace of the input.

At height zero take $P_0 = R_0 = B$ and $C_0 = \lambda_0 = \mathrm{id}_B$. Let $\mathcal{I}$ consist of all pairs (A, J), where A is a line in F^3 and $J : B \to U_A$ is a binary linear isomorphism. Each line has the same number of such isomorphisms.

We include one child for every (A, J) in a single fixed parent map. Choosing a uniform noisy child will then sample a uniform line and a fresh uniform isomorphism. For $t \ge 1$, take $P_t = \prod_{(A,J)\in\mathcal{I}} P_{t-1}$ and set

$$(x, y) = \sum_{A,J} J C_{t-1}(u_{AJ}), \qquad C_t(u) = C_{\mathrm{blk}}(x, y) = y + Q(x). \tag{6.10}$$

A tuple of child shifts changes this aggregate by the sum of their embedded output translations. To make the parent change a pure output translation, we require the X-component of this sum to

vanish. Accordingly, define

$$R_t = \left\{ (h_{AJ}) \in \prod_{A,J} R_{t-1} : \sum_{A,J} J\lambda_{t-1}(h_{AJ}) \in \{0\} \oplus B \right\}, \tag{6.11}$$

and let $\lambda_t(h) = b'$ when this aggregate is $(0, b')$. These are linear definitions, and Equation (6.9) follows immediately from Equation (6.10). Surjectivity follows inductively from Lemma 6.1: the child output translations can be chosen arbitrarily, and their aggregate map is onto $X \oplus B$. Equation (6.9) and surjectivity also show that C_t sends a uniform input to a uniform element of B.

Let ξ_t be the following additive noise in P_t. At height zero it is uniform in B. At a positive height, choose a uniform child (A, J) and embed an independent copy of ξ_{t-1} in that child, with zero in all other children. Equivalently, choose a uniform leaf in the product tree and add a uniform symbol of B at that leaf. Each path choice can be sampled as a uniform A followed by a uniform J, independently at successive levels. All these distributions are rational. Figure 2 illustrates one level of the recursion and the selected noisy branch.

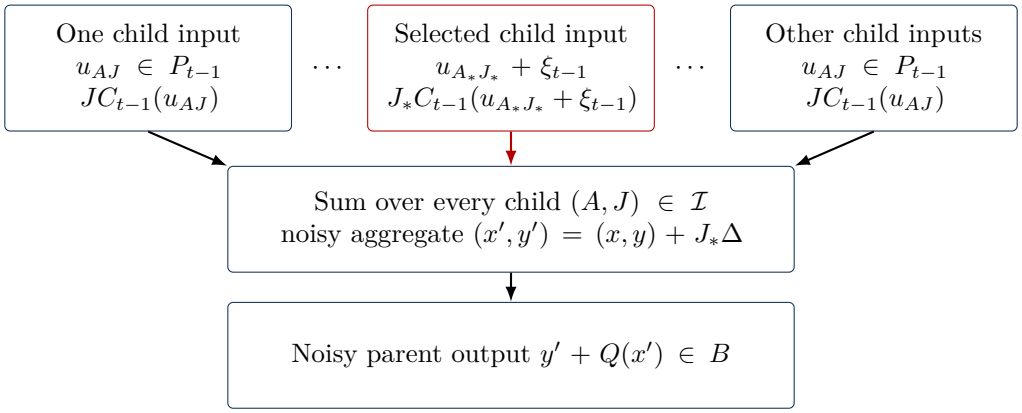


Figure 2: One level of the recursion for the equivariant map, shown under its additive noise. The input space is a product over all child indices (A, J); only one uniformly chosen child is perturbed. Each child output is embedded in $U_A \leq X \oplus B$, all embedded outputs are added, and the quadratic map is applied once at the parent. Here (x, y) is the initial aggregate, and Δ is the selected child's output difference, so the noisy aggregate is $(x', y') = (x, y) + J_*\Delta$. The omitted children obey the same rule. The highlighted perturbation may disappear at the parent.

Lemma 6.6. *For every integer $t \geq 0$, the recursive map C_t and its noise ξ_t satisfy*

$$\Pr[C_t(u + \xi_t) \neq C_t(u)] = (1 - q^{-3})(1 - \theta)^t, \tag{6.12}$$

where u is uniform in P_t and independent of ξ_t.

Proof. Write $p_t = \Pr[C_t(u + \xi_t) \neq C_t(u)]$ under the independent sampling in the statement. The identity at height zero is immediate. Consider one higher level and condition on the selected child (A, J). Write Δ for its output difference. All child copies have the same function and the same inner noise law, so the law of Δ is independent of the selected index.

We need also the conditional law of the child's initial output. For any fixed child perturbation a, Equation (6.9) gives

$$C_{t-1}(u + a + h) + C_{t-1}(u + h) = C_{t-1}(u + a) + C_{t-1}(u) \qquad (h \in R_{t-1}).$$

Thus the difference is invariant under these shifts, while the initial output is translated by every element of B. On uniform input its initial output remains uniform conditional on the difference, even if the inner perturbation is also specified. All other initial child outputs remain independently uniform. Because the aggregate map is onto $X \oplus B$, the initial aggregate (x, y) is therefore uniform conditional on (A, J, Δ). In particular it is independent of the aggregate perturbation $J\Delta$.

Conditional on $\Delta \neq 0$ and A, the isomorphism J remains uniform, because the law of Δ is independent of the selected child index. Thus $J\Delta$ is uniform in $U_A \setminus \{0\}$. The conditional uniformity just proved makes the initial aggregate independent of this update. Lemma 6.2 therefore gives parent change probability $1 - \theta$ conditional on a nonzero child difference. When $\Delta = 0$, the aggregate does not change. Consequently $p_t = p_{t-1}(1 - \theta)$, proving the identity. □

6.4 Linear extensions and loss of rank

The nonlinear change probability now decays with the height. For the fixed-B gadget, our remaining target is detection whenever a family's restrictions fill B^*. We will prove this by tracking every binary-linear choice of extensions of these output linear functionals through the shift action.

For $S \leq B^*$, consider a binary linear family of extensions on P_t, that is, a map $z \mapsto \psi_z$ from S to P_t^* satisfying

$$\psi_z|_{R_t} = z \circ \lambda_t. \tag{6.13}$$

We next describe every such family at a parent.

Lemma 6.7. *Let $t \geq 1$ and let $z \mapsto \psi_z$ be a linear family of extensions on P_t with domain $S \leq B^*$. For each child (A, J), restriction to its copy of R_{t-1} factors through $\lambda_{t-1} : R_{t-1} \to B$. The resulting linear functionals on B are*

$$\phi_{AJ}(z) = J^*\big((\gamma_z, z)|_{U_A}\big) \in B^*, \tag{6.14}$$

where $\gamma : S \to X^*$ is a single binary linear map common to all the children.

Proof. A shift supported on one child and lying in $\ker \lambda_{t-1}$ is a parent shift with output translation zero. Each ψ_z vanishes on it, proving the first assertion.

Let $T = \prod_{A,J} B$ be the space of child output translations, with aggregate map $\mathcal{A} : T \to X \oplus B$. This map is onto. The sum of the child linear functionals, minus z applied to the second component of $\mathcal{A}$, vanishes whenever the first component of $\mathcal{A}$ is zero: every such tuple of translations can be supplied by child shifts, giving a parent shift. The difference therefore factors uniquely through the first component of $\mathcal{A}$, which is onto X. Call this linear functional γ_z. Uniqueness shows that γ_z depends linearly on z. Evaluating on a tuple supported at one child gives Equation (6.14). □

After choosing a child, let $S' = \operatorname{im} \phi_{AJ}$ and choose a linear section $\iota : S' \to S$ of ϕ_{AJ}. Restricting $\psi_{\iota(z')}$ to that child input gives a linear family of extensions with domain S' on P_{t-1}. In fact, if $\jmath_{AJ}$ embeds that child in the parent product, then for $h \in R_{t-1}$,

$$(\psi_{\iota(z')} \circ \jmath_{AJ})(h) = \phi_{AJ}(\iota(z'))(\lambda_{t-1}h) = z'(\lambda_{t-1}h).$$

This checks the shift action required of the new family of extensions even when ϕ_{AJ} has a kernel. It also identifies the surviving linear functional as the restriction of an original linear functional, which is needed when detection is passed back up the selected branch. We may fix deterministic rules for all such sections. In particular they depend only on the already chosen path, and never on a future path index or on the final noise symbol. On descending the tree, every surviving functional is a restriction of a member of the original linear family.

The factorization now connects the recursion to the local restriction problem. For a fixed parent family of extensions, its single map γ is common to every outgoing line A, and $\phi_{AJ} = J^*\kappa_{A,\gamma}$. Since J^* is an isomorphism, it does not change the kernel or rank. We may therefore apply Equations (6.6) and (6.7) when the outgoing line is chosen uniformly. The next proof shows that the incoming isomorphisms make membership in $\mathcal{H}_r$ typical at each positive subsequent rank r, even though the family of extensions itself depends on the earlier path.

Proposition 6.8. *For the field and the bound ε_0 on failure of the two conditions, fixed in Equation (6.8) and the paragraph following it, put*

$$s = \left\lfloor \frac{H_{r_0}}{8\theta} \right\rfloor. \tag{6.15}$$

Every linear family of extensions $z \mapsto \psi_z$ on P_s with domain B^* satisfies

$$\Pr[\textit{there exists } z \in B^* \textit{ with } \psi_z(\xi_s) \neq 0] \geq \frac{1}{4}. \tag{6.16}$$

Proof. Restrict the family with domain B^* at the top to the fixed subspace in $\mathcal{H}_{r_0}$. Follow the random leaf path, replacing the family at each chosen child by the family of extensions on the image just described. Let S_j be its domain after j steps, and let $r_j = \dim S_j$, so initially $r_j = r_0$ at $j = 0$. Continue with the zero space after extinction.

We spell out the conditioning that permits the defining conditions of $\mathcal{H}_r$ to be used at every level. Let $\mathcal{F}_j$ be the information in all path choices before the outgoing choice (A_j, J_j) at step j. The current space, family of extensions, and map γ_j are $\mathcal{F}_j$-measurable. Conditional on $\mathcal{F}_j$ and A_j, the subspace

$$E_{j,A_j} = \operatorname{im}[z \mapsto (\gamma_{j,z}, z)|_{U_{A_j}}] \leq U_{A_j}^*$$

is fixed. The next domain and its rank are

$$S_{j+1} = J_j^* E_{j,A_j}, \qquad r_{j+1} = \dim E_{j,A_j}.$$

The rank is fixed before J_j is revealed. A uniform binary isomorphism $J_j : B \to U_{A_j}$ makes S_{j+1} a uniformly random subspace of B^* of that rank. Therefore, if B_j is the indicator that $r_j > 0$ and $S_j \notin \mathcal{H}_{r_j}$, then

$$\mathbb{E}[B_{j+1} \mid \mathcal{F}_j, A_j] \leq \varepsilon_0. \tag{6.17}$$

The next family of extensions may depend on J_j. This causes no restriction on the argument: membership in $\mathcal{H}_{r_{j+1}}$ bounds every binary linear map g on the selected subspace, including the map supplied by that family.

Let $\Delta_j = H_{r_j} - H_{r_{j+1}} \geq 0$. Conditional on $\mathcal{F}_j$, the outgoing A_j is independent and uniform, and the rank-loss estimates above apply to the fixed current family of extensions. Consequently

$$\mathbb{E}[\Delta_j \mid \mathcal{F}_j] \leq 3\theta + H_{r_0} 2^{r_0} \theta B_j. \tag{6.18}$$

For $j = 0$ the space belongs to $\mathcal{H}_{r_0}$, so its expected drop is at most 3θ. For $j \geq 1$, average Equation (6.18) over the preceding isomorphism using Equation (6.17). This gives the unconditional bound

$$\mathbb{E}\Delta_j \leq (3 + H_{r_0} 2^{r_0} \varepsilon_0)\theta \leq 4\theta.$$

Summing the expected drops and using Equation (6.15),

$$\mathbb{E}[H_{r_0} - H_{r_s}] \leq 4s\theta \leq \frac{H_{r_0}}{2}.$$

Extinction requires a total drop of H_{r_0}, so $\Pr[r_s = 0] \le 1/2$. At a leaf, $P_0 = R_0 = B$ and its family of extensions consists of the linear functionals themselves. Conditional on any surviving rank $r_s > 0$, an independent uniform symbol in B is detected with probability $1 - 2^{-r_s} \ge 1/2$. Every surviving functional is a restriction of the original family along the selected leaf, so such a detection implies a detection by that family of the embedded noise ξ_s. Multiplying the two bounds proves Equation (6.16). □

6.5 Quotients and enlargement

The fixed-B construction is complete once its shift action is realized by an actual subspace. Put

$$V_0 = P_s / \ker \lambda_s,$$

where the kernel is taken inside $R_s \le P_s$, and let $q_0 : P_s \to V_0$ be the quotient map. Equation (6.9) makes C_s invariant under this kernel, so it descends to a map $\overline{C} : V_0 \to B$. The quotient $R_s / \ker \lambda_s$ is identified with B by λ_s, giving an inclusion $i_B : B \to V_0$ such that

$$q_0(h) = i_B(\lambda_s h) \qquad (h \in R_s).$$

We henceforth identify this subspace with B. Its shift action satisfies

$$\overline{C}(x + b') = \overline{C}(x) + b' \qquad (b' \in B).$$

Let ν be the pushforward of the noise law of ξ_s. Uniform input on P_s pushes to uniform input on V_0, so the output change probability remains $(1 - q^{-3})(1 - \theta)^s$.

Every linear subspace $G_0 \le V_0^*$ whose restriction is all of B^* satisfies

$$\Pr_{a \sim \nu}[\text{there exists } g \in G_0 \text{ with } g(a) \ne 0] \ge \frac{1}{4}. \tag{6.19}$$

Indeed, choose a linear section $\sigma : B^* \to G_0$ of restriction and define $\psi_z = q_0^* \sigma(z) = \sigma(z) \circ q_0$. For $h \in R_s$,

$$\psi_z(h) = \sigma(z)(q_0 h) = z(\lambda_s h).$$

Thus this pullback is a linear family of extensions with domain B^*. Moreover, $\psi_z(\xi_s) = \sigma(z)(q_0 \xi_s)$, so detection by the family of extensions is detection by the chosen section on the pushed-forward noise. Proposition 6.8 gives the bound, and detection by the section implies detection by G_0.

Proof of Lemma 3.3. Choose r_0 sufficiently large that

$$\exp(1 - H_{r_0}/8) \le p_*.$$

Then choose ε_0 as in Equation (6.8) and fix d, and hence $b = 3d$ and θ, using Lemma 6.5. Construct the height s in Equation (6.15) and the quotient $(V_0, B, \overline{C}, \nu)$ above. Lemma 6.6 gives

$$(1 - q^{-3})(1 - \theta)^s \le (1 - \theta)^s \le \exp(-\theta s) \le \exp(\theta - H_{r_0}/8) \le p_*. \tag{6.20}$$

Set $r_* = b + 2$. All choices so far depend only on p_*.

We now enlarge the output space. The point of using random injections $B \to K$ is that a family of restriction rank at least $b + 2$ on K restricts onto all of B^* along most injections. A copy of the base gadget for each injection will therefore turn its full-restriction detection bound into the required high-rank bound.

Fix $\ell \geq b+2$ and write $K = \mathbb{F}_2^\ell$. Let $\mathcal{L}$ be the finite set of all binary linear injections $L : B \to K$. Take a product $P = \prod_{L\in\mathcal{L}} V_0$ and define

$$\widetilde{C}((x_L)_L) = \sum_{L\in\mathcal{L}} L\overline{C}(x_L).$$

The product of shift subspaces $T = \prod_L B \leq P$ acts on this output through the linear map

$$\Lambda : T \longrightarrow K, \qquad \Lambda((b_L)_L) = \sum_L Lb_L.$$

This map is onto, because every nonzero vector of K is in the image of some injection $B \to K$. Quotient by its kernel inside T:

$$\mathcal{V} = P/\ker\Lambda.$$

The map $\widetilde{C}$ descends to $C : \mathcal{V} \to K$, and $T/\ker\Lambda$ identifies with the required shift subspace K. This proves Equation (3.3).

On P, choose a uniform $L \in \mathcal{L}$ and update only that copy by an independent sample from ν. Let μ be the resulting law after the quotient. On uniform product input the output sum changes exactly when the selected copy's output changes, because L is injective. Uniform input pushes to uniform input on the quotient, so the change probability is still $(1-q^{-3})(1-\theta)^s$. Equation (6.20) proves Equation (3.4). Every noise distribution used here is obtained from uniform finite choices by pushforward, and therefore is rational.

It remains to establish detection for an arbitrary $G \leq \mathcal{V}^*$. Let $S = G|_K \leq K^*$, of dimension $r \geq r_*$. Pull G back to P and then restrict to its copy of V_0 indexed by L; denote the resulting subspace of V_0^* by G_L. Its restriction to that copy's shift subspace B is exactly

$$G_L|_B = L^*S.$$

To see this, a shift $b' \in B$ supported on copy L represents the shift $Lb' \in K$ in the quotient, so a linear functional restricting to $z \in S$ evaluates there as $z(Lb')$.

The map $L^* : S \to B^*$ fails to be onto if and only if there is a nonzero $b' \in B$ with $Lb' \in S^\perp$. For each such fixed b', a uniform injection L makes Lb' uniform among the nonzero vectors of K. Hence

$$\Pr_L[Lb' \in S^\perp] = \frac{2^{\ell-r}-1}{2^\ell-1} \leq 2^{-r}.$$

A union bound gives

$$\Pr_L[L^*S \neq B^*] \leq (2^b-1)2^{-r} \leq 2^{b-r} \leq \frac{1}{4}.$$

On each of the remaining copies, Equation (6.19) gives detection probability at least $1/4$ by G_L, and thus by G on the embedded noise. Averaging over the selected copy yields detection probability at least $\frac{3}{4}\cdot\frac{1}{4} = \frac{3}{16} \geq \frac{1}{8}$, proving Equation (3.5). The construction works for every $\ell \geq b+2$, with the already fixed $r_* = b+2$, as required.

For completeness, all choices are effective when p_* is rational. Choose an integer $m \geq 0$ with $2^{-m} \leq p_*$ and then choose r_0 with $H_{r_0} \geq 8(m+1)$; since $e^{-m} \leq 2^{-m}$, this ensures the initial bound on r_0. Choose rational ε_0 satisfying Equation (6.8). For successive integers d with $3d \geq r_0$, enumerate the binary subspaces of $\mathbb{F}_{2^d}^3$ and the linear maps in the definition of $\mathcal{H}_r$. This tests membership exactly and computes the fraction of subspaces outside $\mathcal{H}_r$ at each rank $r = 1, \ldots, r_0$. Lemma 6.5 ensures that eventually all these fractions are at most ε_0. Fix the first such field and a subspace in $\mathcal{H}_{r_0}$. The remaining products, isomorphisms, injections, quotients, function tables, and rational noise laws are obtained by finite enumeration and binary linear algebra. This gives the asserted terminating construction. □

7 Matrix extraction and Fourier decoding

Section 7.1 proves Theorem 4.1. The following subsections prove Proposition 4.3, with the parameters, table family, and advice law of Section 4, and Section 7.6 closes with the search for the tuple length. Lemma 7.1 first converts acceptance under the noise law μ into acceptance under uniform rank-one noise. The inverse theorem then supplies slices, and Lemma 7.2 makes slices with the required agreement occur on a positive fraction of the visible row advice. Lemma 7.4 transfers a chosen slice to the projected question. Finally, Lemma 7.5 converts that agreement into the probability of agreement of the two private decoders.

7.1 Proof of the shortcode soundness theorem

We recall what Theorem 4.1 asserts. Given $\eta \in (0,1)$, we must find $\alpha \in (0,1]$, an integer $r_{\mathrm{s}} \geq 1$ and an integer ℓ_0, depending only on η, and for each $\ell \geq \ell_0$ a threshold $m_0(\eta,\ell)$, such that the following holds for every $m \geq m_0$. Every function $f : \mathrm{Mat}_{\ell,m} \to \mathbb{F}_2^{\ell}$ whose equality acceptance under the uniform rank-one step is at least η, as in Equation (4.2) with $M \in \mathrm{Mat}_{\ell,m}$, $a \in \mathbb{F}_2^{\ell}$ and $l \in \mathbb{F}_2^{m}$ independent and uniform, agrees with some affine evaluation $Mz + u$ on at least an α fraction of some nonempty row and column slice (4.1) with $n_r, n_c \leq r_{\mathrm{s}}$, as in Equation (4.3). No folding condition is assumed, and either factor of the step may be zero. The proof ends by making these choices effective, with rational α for rational η, as asserted after the theorem.

Proof of Theorem 4.1. We derive this consequence from the Grassmann expansion theorem of Khot–Minzer–Safra [25, Theorem 1.12 in the 2018 revision], using the matrix chart of Barak–Kothari–Steurer [6, Definition 3.2 and Lemmas 3.3–3.7].

The Grassmann graph on an ambient binary space E has the ℓ-dimensional subspaces as vertices; two vertices are adjacent when their intersection has dimension $\ell - 1$. For a nonempty vertex set, consider the probability that a uniform vertex in the set has a uniform neighbor still in the set. The cited theorem says that a set with this probability at least a fixed $\zeta > 0$ has positive relative density in some interval

$$\{L : A_0 \subseteq L \subseteq B_0\}, \qquad \dim A_0 + \operatorname{codim}_E B_0 \leq r.$$

The density bound and r depend only on ζ. The theorem holds for all sufficiently large ℓ, and then all sufficiently large $\dim E$.

Equality acceptance is the average of the probabilities of remaining in the nonempty fibers $f^{-1}(y)$ under the matrix step, weighted by their uniform masses. Some fiber S therefore has probability at least η of remaining in S after one step. The probability that either perturbation factor is zero is

$$p_0 = 2^{-\ell} + 2^{-m} - 2^{-\ell-m}.$$

Choose the dimension thresholds so that $p_0 \leq \eta/2$. Conditional on both factors being nonzero, the probability of remaining in S is at least $(\eta - p_0)/(1 - p_0) \geq \eta/2$.

For each matrix put

$$L_M = \{(x, M^{\top}x) : x \in \mathbb{F}_2^{\ell}\} \subseteq E := \mathbb{F}_2^{\ell} \oplus \mathbb{F}_2^{m}.$$

The map $M \mapsto L_M$ is a bijection onto the chart $\mathcal{C}$ of subspaces whose projection onto the first summand is invertible. Since

$$\dim(L_M \cap L_N) = \ell - \operatorname{rank}(M - N),$$

the neighbors within the chart correspond exactly to nonzero rank-one matrix differences. Each chart vertex has $(2^\ell-1)(2^m-1)$ such neighbors, whereas its full Grassmann degree is $(2^\ell-1)(2^{m+1}-2)$: choose a hyperplane in the vertex and then a one-dimensional extension in the quotient by that hyperplane, other than the vertex itself. Thus exactly half of its neighbors lie in the chart. Over $\mathbb{F}_2$, every nonzero rank-one matrix has a unique factorization $al^\top$ with nonzero factors. The lifted set $\widetilde{S} = \{L_M : M \in S\}$ consequently has probability at least $\eta/4$ of remaining in $\widetilde{S}$ after one Grassmann step.

Apply the expansion theorem with $\zeta = \eta/4$, obtaining density α on an interval with $\dim A_0 + \operatorname{codim}_E B_0 \le r_{\mathrm{s}}$. Intersecting this interval with $\mathcal{C}$ only increases the relative density of $\widetilde{S}$, since $\widetilde{S} \subseteq \mathcal{C}$, and the intersection is nonempty. A basis (d_i, s_i) of A_0 specifies $A_0 \subseteq L_M$ by the equations $d_i^\top M = s_i^\top$. For the standard dot product on E, a basis (t_j, q_j) of $B_0^\perp$ specifies $L_M \subseteq B_0$ by $Mq_j = t_j$. Their simultaneous solution set is a slice of the required form, with at most r_{s} equations of either kind. On at least an α fraction of this slice, f equals the value defining S. This proves the conclusion with $z = 0$ and u equal to that value.

The constants α, r_{s} were chosen using only η. Since $\dim E = \ell + m$, after fixing ℓ the remaining dimension threshold is absorbed into $m_0(\eta, \ell)$. No folding condition was used. The fiber argument is the equality consequence in [6, Lemma 2.3]; numbered references to that paper use its arXiv version 1.

For the effective assertion, put $\zeta = \eta/4$. The quantitative argument in [25, Section 2.4, Theorem 2.13, and Lemmas 2.7–2.8] gives the following bound. If a nonempty set Y of vertices of the Grassmann graph in ambient dimension n has relative density at most α on every interval with $\dim A_0 + \operatorname{codim} B_0 \le r \le \ell$, then its probability of remaining in the set is at most

$$(r+1)2^{7r^3+3}(\alpha + 2^{\ell+r-n})^{1/4} + 2^{-(r+1)} + 2^{-\ell} + 2^{\ell-n+1}. \tag{7.1}$$

To obtain this expression, lift Y to the set Y^* of all ordered bases of its members, a basis-invariant set in the source's graph $H_{n,\ell}$. This graph has vertex space $(\mathbb{F}_2^n)^\ell$ and step $(x_i)_{i=1}^\ell \mapsto (x_i + b_i y)_{i=1}^\ell$, with independent uniform $b \in \mathbb{F}_2^\ell$ and $y \in \mathbb{F}_2^n$. The hypothesis says that Y is (r, α)-pseudorandom in the sense of [25, Definition 2.3], so Lemma 2.8 there makes Y^* $(r, \alpha + 2^{\ell+r-n})$-pseudorandom in the sense of [25, Definition 2.5]. This means that fixing any q coordinates and sampling the remaining coordinates independently and uniformly from any subspace W_0 with $q + \operatorname{codim} W_0 \le r$ gives membership probability at most $\alpha + 2^{\ell+r-n}$.

Let F be the indicator of Y^*, let $\delta > 0$ be its density, and let $F_{=i}$ be the part of the Fourier expansion of F on the characters whose ℓ frequency vectors span a space of dimension i; these characters have eigenvalue 2^{-i} under the step [25, Lemma 2.10]. Theorem 2.13 there gives $\|F_{=i}\|_2^2 \le 2^{7r^3+3}(\alpha + 2^{\ell+r-n})^{1/4}\delta$ for $0 \le i \le r$. Parseval gives $\sum_{i=0}^{\ell} \|F_{=i}\|_2^2 = \delta$. Hence, as in [25, Section 2.4], the probability ρ^* of remaining in Y^* after one step in $H_{n,\ell}$ satisfies

$$\begin{aligned}\delta\rho^* &= \sum_i 2^{-i}\|F_{=i}\|_2^2 \le \sum_{i \le r} \|F_{=i}\|_2^2 + 2^{-(r+1)} \sum_{i>r} \|F_{=i}\|_2^2 \\ &\le (r+1)2^{7r^3+3}(\alpha + 2^{\ell+r-n})^{1/4}\delta + 2^{-(r+1)}\delta.\end{aligned}$$

Divide by δ. Lemma 2.7 there bounds the difference between the probabilities of remaining in Y^* and Y after one step by $2^{-\ell} + 2^{\ell-n+1}$; adding it returns to the Grassmann graph and gives Equation (7.1).

Choose $r \ge 1$ with $2^{-(r+1)} \le \zeta/4$, and then a positive rational $\alpha \le 1$ with

$$2\alpha\big((r+1)2^{7r^3+3}\big)^4 \le (\zeta/4)^4.$$

Take $\ell \geq \max\{2, r\}$ with $2^{-\ell} \leq \zeta/8$ and, after fixing ℓ, take m so large that $2^{r-m} \leq \alpha$ and $2^{1-m} \leq \zeta/8$. With $n = \ell + m$, Equation (7.1) is at most $3\zeta/4 < \zeta$. Thus a set with probability at least ζ of remaining in the set after one step must have density greater than α on one of the required intervals. These same bounds make $p_0 \leq \eta/2$. They give effective choices $r_{\mathrm{s}} = r$, ℓ_0, and $m_0(\eta, \ell)$ for the preceding argument, with the required order of dependence. □

The next two subsections work for any positive integer k, with a fixed family satisfying Equation (4.6). They use unrestricted uniform tables on U. We specialize k and β when comparing that law with the projected experiment.

7.2 Comparing the two matrix perturbations

The uniform matrix perturbation has the rank-indexed Fourier spectrum computed in [25, Lemma 2.10 in the 2018 revision]. We compare that spectrum with the noise law μ.

Lemma 7.1 (Spectral comparison)**.** *If the table family passes Equation* (3.9) *with probability at least* 0.99*, then*

$$\mathbb{E}_{U,T} \Pr_{M,a,l}[F_U(M, T) = F_U(M + al^{\top}, T)] \geq 0.9 \cdot 2^{-r_*}. \tag{7.2}$$

Here $U \sim \omega^{\otimes k}$; conditional on U, the maps $T \in \mathrm{Hom}(L_U, W)$ and $M \in \mathrm{Hom}(L_U, K)$ and the vectors $a \in K$, $l \in \mathbb{F}_2^m$ are independent and uniform.

Proof. For $\sigma \in K^*$ define the sign function $f_{U,\sigma}(P) = (-1)^{\sigma(F_U(P))}$. A frequency on the matrix domain is a tuple $g = (g_i)_{i=0}^{m-1}$ of functionals in $\mathcal{V}^*$; its character at P is $(-1)^{\sum_i g_i(Pe_i)}$, where $e_0, \ldots, e_{m-1}$ are the coordinate vectors of L_U. All Fourier coefficients use uniform probability measure. Translation by $al^{\top}$ multiplies this character by $(-1)^{\sum_i l_i g_i(a)}$. Averaging over uniform l therefore gives the eigenvalue

$$\Lambda_g = \Pr_{a \sim \mu}[g_i(a) = 0 \text{ for every } i].$$

It lies in $[0, 1]$. By Lemma 3.3, it is at most $7/8$ whenever

$$\dim \operatorname{span}\{g_i|_K : 0 \leq i < m\} \geq r_*.$$

Indeed the span of the g_i is a subspace of $\mathcal{V}^*$ to which the lemma applies.

Expanding equality by a uniform output character, and then using Fourier orthogonality, expresses the test acceptance as

$$\mathbb{E}_{U,\sigma} \sum_g |\widehat{f}_{U,\sigma}(g)|^2 \Lambda_g.$$

Parseval gives total squared mass one for each sign function. If B is the average squared mass on restricted ranks at least r_*, then $0.99 \leq 1 - B/8$, so $B \leq 0.08$. In particular, at least 0.9 of the average mass lies on ranks below r_*.

When a is instead uniform in K, the eigenvalue at g is $2^{-\dim \operatorname{span}\{g_i|_K\}}$. The preceding mass thus gives the lower bound in Equation (7.2). This perturbation leaves T fixed, so the expression there is precisely the average of the ordinary shortcode equality tests at fixed (U, T). □

7.3 Agreement on a positive fraction of row fibers

The inverse theorem gives one structured slice on each qualifying (U, T) fiber. The U-prover's information about the K-valued matrix is the pair (A, AM). We need a positive probability that these observed rows determine a slice with the required agreement. We will also identify the short set of valid answers supplied by that slice. The erasure argument below proves the probability bound, and folding then makes the answer set nonempty.

For this extraction step, sample U as before and take independent uniform maps $T : L_U \to W$, $A : K \to \mathbb{F}_2^{r_s}$, and $M : L_U \to K$. For fixed U, T, A and a map $S_0 : L_U \to \mathbb{F}_2^{r_s}$, the set $\{M : AM = S_0\}$ is its *row fiber*. Let $\mathcal{R}$ consist of the row advice (U, A, T, S_0) for which there exist at most r_s column specifications and a target $Mz+u$ such that the simultaneous solution set of the column specifications and $AM = S_0$ is nonempty and $F_U(M, T) = Mz+u$ with probability at least $\alpha/2$ under the uniform measure on this set.

This is an analytic condition on the row fiber. The target vector z need not initially be an actual answer. The fixed column values allow z to vary modulo the span of their column directions, with a corresponding change in u. Thus the slice naturally supplies a coset of possible target vectors. Its valid part will be the answer set sampled by the U-prover.

Lemma 7.2 (Row advice admitting agreement and its short answer set)**.** *Fix the parameters of Equation (4.4) and a table family satisfying Equation (4.6). For every sufficiently large positive integer k, acceptance at least 0.99 in the matrix test implies*

$$\Pr_{U,T,A,M}[(U, A, T, AM) \in \mathcal{R}] \geq g_0, \qquad g_0 = 10\eta^2\, 2^{-\ell r_s} > 0. \tag{7.3}$$

The lower bound and the required largeness of k are independent of the parity instance and the family.

For every positive integer k, each row advice in $\mathcal{R}$ admits a deterministically chosen certificate consisting of at most r_s column specifications $Mq_j = t_j$, with $q_j \in L_U$ and $t_j \in K$, and a target $Mz + u$, with $z \in L_U$, $u \in K$, and $e_0^\top z = 1$, satisfying the definition of $\mathcal{R}$. Writing $Z = \operatorname{span}\{q_j\}$, the chosen witness determines the answer set

$$\mathcal{A} = (z + Z) \cap \{v \in L_U : e_0^\top v = 1\}, \qquad 1 \leq |\mathcal{A}| \leq 2^{r_s}. \tag{7.4}$$

The witness and $\mathcal{A}$ depend only on (U, A, T, S_0) and the fixed table family. The slice and its target retain the agreement guarantee in the definition of $\mathcal{R}$.

Proof. By Lemma 7.1, the average shortcode equality acceptance is at least $0.9 \cdot 2^{-r_*} = 14.4\eta$. If q is the probability that the acceptance at (U, T) is at least 4η, the average is at most $4\eta + q$, so $q \geq 10.4\eta \geq 10\eta$. Fix such a pair (U, T) and write $f(M) = F_U(M, T)$. Let B be the union, in the matrix domain, of all row fibers of advice in $\mathcal{R}$, over every possible A, S_0. For each $M \in B$, at least one of the $2^{\ell r_s}$ maps A gives advice in $\mathcal{R}$ on the row fiber containing M, namely the fiber of $S_0 = AM$. The uniform choice of A hits that map with probability $2^{-\ell r_s}$, so at this (U, T) the probability over A, M that $(U, A, T, AM) \in \mathcal{R}$ is at least $2^{-\ell r_s} \Pr[B]$. It therefore suffices to show that $\Pr[B] \geq \eta$. We will erase all row fibers of advice in $\mathcal{R}$. If their union B were too small, acceptance would survive the erasure, while independent random replacement values would destroy every possible agreement certificate. Suppose that its uniform measure is less than η. Change f at the entries of B, replacing each by an independent uniform element of K. Each matrix receives one replacement, even when several such row fibers overlap there. Both endpoints of the shortcode step are uniform matrices. Consequently every such modification changes acceptance by at most $2 \Pr[B]$, and the modified function has acceptance at least 2η.

Put $r = r_{\mathrm{s}}$. We claim that the replacement values can be chosen so that the following holds simultaneously for every nonempty slice $\mathcal{S}$ with at most r row and r column specifications and every target $Mz + u$, with $z \in \mathbb{F}_2^m$ and $u \in K$: fewer than $(\alpha/4)|\mathcal{S}|$ matrices $M \in \mathcal{S} \cap B$ receive $Mz + u$ as their replacement value. The space of differences of matrices in such a slice is

$$\operatorname{Hom}\bigl(\mathbb{F}_2^m / \operatorname{span}\{q_j\},\ \bigcap_i \ker d_i^\top\bigr).$$

Its size $n_{\mathcal{S}}$ is therefore at least $2^{(\ell-r)(m-r)}$. Padding the lists of specifications with zero equations, the number of slice–target descriptions is at most $2^{(2r+1)(\ell+m)}$.

For a fixed description, count only matches at randomized entries and assign a deterministic zero contribution to the other entries. These are independent variables in $[0, 1]$, with total mean at most $2^{-\ell} n_{\mathcal{S}} < \alpha n_{\mathcal{S}}/8$. Hoeffding's inequality [19, Theorem 2, equation (2.6)] bounds the probability of at least $\alpha n_{\mathcal{S}}/4$ such matches by $\exp(-\alpha^2 n_{\mathcal{S}}/32)$: apply it to the average of $n_{\mathcal{S}}$ summands with deviation at least $\alpha/8$, giving exponent $-2n_{\mathcal{S}}(\alpha/8)^2$. The union bound over all descriptions is thus at most

$$\exp\left((2r+1)(\ell+m)\log 2 - \frac{\alpha^2}{32} 2^{(\ell-r)(m-r)}\right),$$

which tends to zero as $m = 1 + 2k$ grows. Fix replacements with the asserted property.

Consider any slice of the allowed type. If its row advice was in $\mathcal{R}$, every entry of that fiber belongs to B, and its modified agreement is less than $\alpha/4$. If the row advice was outside $\mathcal{R}$, the original agreement with each target on the slice was less than $\alpha/2$. The unmodified matches form a subset of those original matches, and the randomized matches add less than $\alpha/4$. The modified agreement is then less than $3\alpha/4$. Padding shorter row systems by zeros shows that these two cases cover every slice in Theorem 4.1. That theorem now contradicts the modified function's acceptance at least 2η. Its applicability does not require the modification to preserve folding.

It follows that $\Pr[B] \geq \eta$, so at this fixed (U, T) the probability that $(U, A, T, AM) \in \mathcal{R}$ is at least $\eta 2^{-\ell r_{\mathrm{s}}}$. At least 10η of the pairs qualify, as shown at the start of the proof, proving Equation (7.3). All counting and concentration bounds depended only on the fixed parameters and m, as required.

The remaining assertion does not use high test acceptance or largeness of k. Fix any positive integer k, a family satisfying Equation (4.6), and any row advice in $\mathcal{R}$. Fix a witness, write its column specifications as $Mq_j = t_j$, and put $Z = \operatorname{span}\{q_j\}$. Nonemptiness of the slice makes the prescribed values consistent: they determine a linear map $t : Z \to K$ with $t(w) = Mw$ throughout the slice. Replacing z by $z + w$ and u by $u + t(w)$ for $w \in Z$ therefore preserves the target. It suffices to show that the coset $z + Z$ contains a vector of first bit one.

Otherwise z and every vector in Z have first bit zero. Put $H = \ker A$. For each $h \in H$, translation $M \mapsto M + he_0^\top$ preserves the row fiber, the column specifications, and the target. By folding it translates $F_U(M, T)$ by h. The action is free, and in each of its H-orbits at most one matrix can satisfy the target equality. The agreement is at most

$$|H|^{-1} = 2^{-\ell + \operatorname{rank} A} \leq 2^{-(\ell - r_{\mathrm{s}})} < \alpha/8,$$

contradicting membership in $\mathcal{R}$. Choose $w \in Z$ for which $z + w$ has first bit one and make the corresponding changes to z and u. They preserve the original agreement fraction. Choose a witness and such a representative using a fixed order on the finite possibilities. This choice uses only the row advice and the fixed table family. Its vector z lies in $\mathcal{A}$, so the set is nonempty; since $\dim Z \leq r_{\mathrm{s}}$, its size is at most $2^{r_{\mathrm{s}}}$. □

7.4 Transferring the chosen slice

The comparison with sparsely projected questions follows the smooth outer-game method of Khot–Safra [28, Section 3 and Lemma 3.1], with shared linear advice as in Khot–Minzer–Safra [26, Sections 3.3–3.4]. Lemma 7.2 gives a positive probability of row advice that determines a slice and a short set of valid answers under unrestricted sampling on U. In the actual experiment, the maps are sampled on O and padded to U. We compare the complete padded table with the unrestricted table, including the affine intercept, with projection choices averaged out, and then transfer the chosen slice to O.

From now on take k along integer cubes larger than one and set $\beta = k^{-2/3}$ in Definition 4.2.

The choice $\beta = k^{-2/3}$ serves two purposes: $k\beta^2 \to 0$ makes the padded table law approach the unrestricted law in Lemma 7.3, while $k\beta \to \infty$ makes the expected number of singleton coordinates diverge. Once the advice spaces are fixed, each singleton has a uniformly positive probability that every visible advice row has zero coefficient on its variable bit. Lemma 4.4 uses these coordinates to bound agreement.

Lemma 7.3 (The padded matrix is asymptotically uniform)**.** *For each equation question U, the padded table*

$$(M, T) = (M_O\bar{\pi}, T_O\bar{\pi})$$

in Definition 4.2, with $\beta = k^{-2/3}$, has conditional distribution at total variation distance $o(1)$ from the unrestricted uniform distribution on $\mathrm{Hom}(L_U, K) \times \mathrm{Hom}(L_U, W)$, as k tends to infinity along cubes. The bound is uniform in U and the parity instance. The projection choices are averaged out in this comparison.

Proof. Combine the two components into a $\mathcal{V}$-valued table and put $n = |\mathcal{V}|$. On an unprojected equation block the two slopes are independent uniform elements of $\mathcal{V}$. On a singleton block the padded pair of slopes is an equal mixture of

$$(a, 0), \qquad (0, a), \qquad (a, a), \qquad a \text{ uniform in } \mathcal{V}.$$

Call these pair laws P_0 and Q_0, respectively. The likelihood ratio dQ_0/dP_0 is n at $(0, 0)$, $n/3$ at the other $3(n-1)$ points on the three displayed lines, and zero elsewhere. Thus

$$\chi^2(Q_0 \| P_0) := \mathbb{E}_{P_0}\left[\left(\frac{dQ_0}{dP_0} - 1\right)^2\right] = \frac{n-1}{3}.$$

The one-block mixture $(1-\beta)P_0 + \beta Q_0$ consequently has chi-squared divergence $\beta^2(n-1)/3$ from P_0. Blocks are sampled independently, including when their question IDs coincide. Multiplying their likelihood ratios gives divergence

$$\left(1 + \frac{n-1}{3}\beta^2\right)^k - 1 = o(1), \tag{7.5}$$

because n is fixed and $k\beta^2 = k^{-1/3}$ tends to zero. Total variation is at most one half the square root of this expression.

For a projection onto the third bit of equation e, the padded intercept acquires the term $b_e a$, where a is that singleton's slope. The original intercept is independent and uniform in $\mathcal{V}$. Adding the sum of these terms leaves it uniform and independent even after all projection choices and slopes are given. The preceding slope estimate therefore proves the assertion for the entire table. Neither the slope laws nor the bound in Equation (7.5) depend on the equation IDs or their right-hand sides. □

Fix U-advice in $\mathcal{R}$ for a moment, with its chosen witness and answer set $\mathcal{A}$ from Lemma 7.2. For a coordinate projection put $z' = \bar{\pi}z$ and $Z' = \bar{\pi}Z$. Since $\bar{\pi}$ preserves the first coordinate,

$$\bar{\pi}\mathcal{A} = (z' + Z') \cap \{v \in L_O : e_0^\top v = 1\}. \tag{7.6}$$

Indeed, every vector in the coset on the right has a preimage in $z + Z$, and the preimage has the same first coordinate. We will show that the O-prover's private Fourier sample reaches this projected answer set.

The certifying slice does not automatically survive projection. For example, a column condition $Mq = t$ becomes $M_O\bar{\pi}q = t$; it is inconsistent if $\bar{\pi}q = 0$ but $t \neq 0$. More generally, projected column directions may acquire new dependencies. The next argument uses the comparison of the full table laws to find a positive probability of samples where the chosen slice remains nonempty and retains enough agreement.

For this analysis, write $\mathsf{D} = (U, O, \pi, A, T_O, S_O)$ for the joint data that determine both prover messages and the projection. These data do not specify M_O beyond the constraint $AM_O = S_O$.

Lemma 7.4 (Transfer of a chosen witness)**.** *Fix a table family satisfying Equation (4.6) and having matrix-test acceptance at least 0.99. In the projected advice experiment with $\beta = k^{-2/3}$, for every sufficiently large integer cube k, there is a set of values of the joint data D, of total probability at least*

$$\gamma_1 = \frac{\alpha g_0}{8}\, 2^{-\ell r_{\mathrm{s}}} > 0 \tag{7.7}$$

such that every D in this set has the following properties. The padded U-advice belongs to $\mathcal{R}$, with the chosen answer set $\mathcal{A}$ and its certifying witness. Writing $z' = \bar{\pi}z$, that witness transfers to a nonempty slice

$$AM_O = S_O, \qquad M_O\bar{\pi}q_j = t_j,$$

on which

$$\Pr\big[F_O(M_O, T_O) = M_O z' + u \mid AM_O = S_O,\ M_O\bar{\pi}q_j = t_j \text{ for all } j\big] \geq \alpha/4. \tag{7.8}$$

The probability in this equation is taken over a uniform matrix in the specified slice.

Proof. First use unrestricted sampling of U, A, M, T, setting $S_0 = AM$. Let $\mathcal{J}$ be the event that $(U, A, T, S_0) \in \mathcal{R}$ and M satisfies its chosen column specifications. Let $\mathcal{M}$ be equality with the chosen target, declaring this event false for row advice outside $\mathcal{R}$. Within a row fiber of advice in $\mathcal{R}$ the nonempty column slice imposes at most ℓr_{s} additional scalar linear equations. Hence Lemma 7.2 gives

$$\Pr[\mathcal{J}] \geq g_0\, 2^{-\ell r_{\mathrm{s}}}, \qquad \Pr[\mathcal{J} \cap \mathcal{M}] \geq (\alpha/2)\Pr[\mathcal{J}].$$

It follows that

$$\Pr[\mathcal{J} \cap \mathcal{M}] - (\alpha/4)\Pr[\mathcal{J}] \geq \frac{\alpha g_0}{4}\, 2^{-\ell r_{\mathrm{s}}}. \tag{7.9}$$

Both events in this expression depend only on (U, A, M, T). Adding the independent A to Lemma 7.3 preserves its variation bound. We may therefore compare the two event probabilities with their values in the padded experiment. If the total variation distance is $v_k = o(1)$, the difference in Equation (7.9) changes by at most $(1 + \alpha/4)v_k$. For all sufficiently large k the padded difference is at least γ_1.

Now condition on D. The remaining M_O is uniform on the row fiber $AM_O = S_O$. The padded U-advice, and therefore its chosen witness and answer set, are fixed. On data with U-advice in $\mathcal{R}$ let p_{D} be the conditional probability of the projected column slice, and, when $p_{\mathsf{D}} > 0$, let q_{D} be its

conditional equality fraction. Put $p_{\mathsf{D}} = 0$ on data with U-advice outside $\mathcal{R}$, and put $q_{\mathsf{D}} = 0$ when $p_{\mathsf{D}} = 0$. Sharing gives $F_U(M_O\bar{\pi}, T_O\bar{\pi}) = F_O(M_O, T_O)$, so the padded difference is exactly

$$\mathbb{E}_{\mathsf{D}}\big[p_{\mathsf{D}}(q_{\mathsf{D}} - \alpha/4)\big] \geq \gamma_1.$$

The integrand is nonpositive outside the event consisting of U-advice in $\mathcal{R}$, $p_{\mathsf{D}} > 0$, and $q_{\mathsf{D}} \geq \alpha/4$, and it is at most one everywhere. That event thus has probability at least γ_1 and has precisely the asserted properties. The law comparison applied to events depending only on (U, A, M, T); conditioning on D here only identifies where their positive difference occurs in the projected experiment. □

7.5 Decoding the row advice

Extracting an outer strategy from Fourier mass follows the approach of Håstad [18, Section 1.1] and the Fourier-list method of Khot–Safra [28, Section 1.1]. Here the projected prover samples squared Fourier mass on its own row fiber. The transferred slice will show that this private sample reaches the projected answer set $\bar{\pi}\mathcal{A}$ with positive probability.

We now define both strategies from their separate messages. The joint data and the transferred witness serve only in the analysis of their agreement; the O-prover does not use the U-prover's witness or answer set.

The U-prover uses its visible row advice to make the deterministic choice in Lemma 7.2. When the advice belongs to $\mathcal{R}$, it samples a_U uniformly from $\mathcal{A}$. This is a nonempty set of at most $2^{r_{\mathrm{s}}}$ actual answers. On other inputs the prover returns any actual answer.

The O-prover starts from its own row fiber. Given (O, A, T_O, S_O), put $H = \ker A$ and choose, by a fixed rule, a representative M_0 satisfying $AM_0 = S_O$. Such a representative exists on every input in the experiment. Write the row fiber as $M_0 + \mathcal{N}$, where $\mathcal{N} = \mathrm{Hom}(L_O, H)$. Sample uniform $\sigma \in K^*$ and form the sign function

$$f_\sigma(N) = (-1)^{\sigma(F_O(M_0+N, T_O))}, \qquad N \in \mathcal{N}.$$

Sample a Fourier frequency $\Phi \in \mathcal{N}^*$ with probability $|\widehat{f_\sigma}(\Phi)|^2$. Identify $\mathcal{N}^* = H^* \otimes L_O$, with the pairing defined on pure tensors by

$$\langle \varphi \otimes v, N \rangle = \varphi(Nv).$$

For $h \in H$, contraction sends Φ to a vector $\Phi(h) \in L_O$. If $\sigma|_H \neq 0$, choose by a fixed rule an $h \in H$ with $\sigma(h) = 1$ and return $\Phi(h)$. If the restriction is trivial, return any actual answer.

This rule always returns an actual answer at frequencies of positive mass. Indeed the folding identity in Equation (4.6) gives

$$f_\sigma(N + he_0^\top) = -f_\sigma(N) \qquad \text{when } \sigma(h) = 1.$$

Changing variables in a Fourier coefficient shows that its being nonzero requires $(-1)^{\langle \Phi, he_0^\top \rangle} = -1$, equivalently $e_0^\top \Phi(h) = 1$.

Figure 3 summarizes the information and private choices of the two decoders.

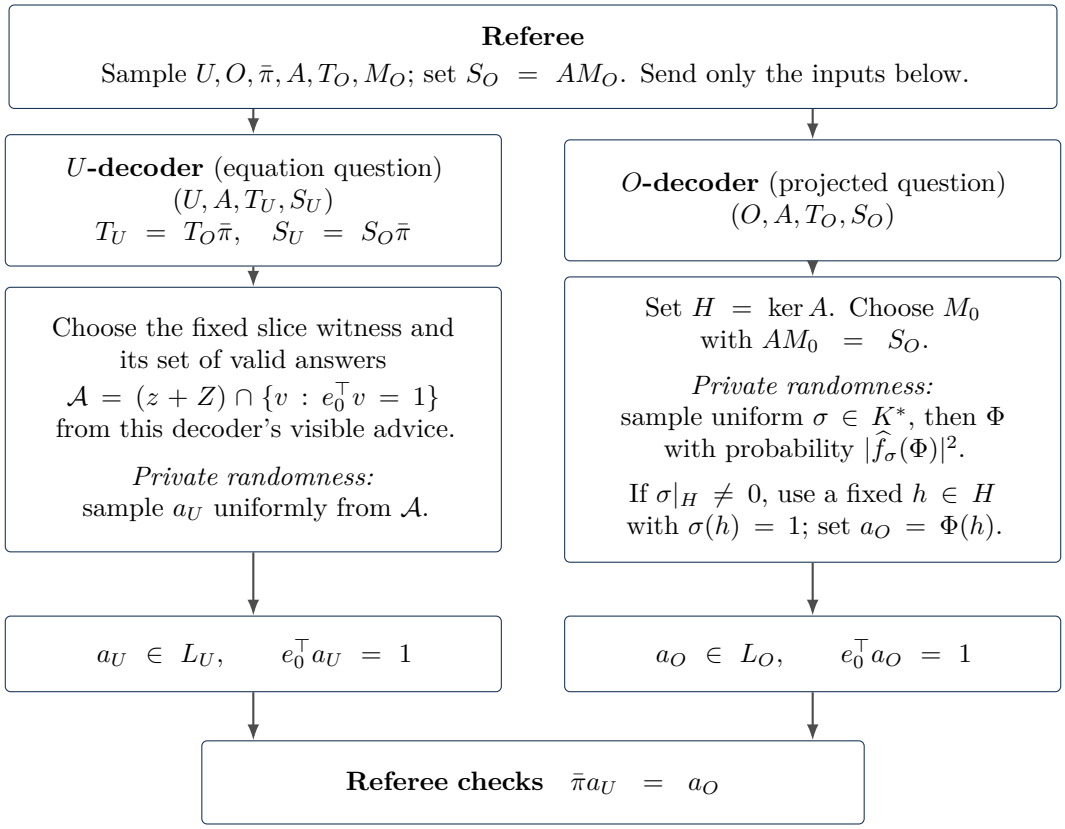


Figure 3: The two decoders use independent random tapes. Neither receives the sampled M_O or the projection $\bar{\pi}$; the O-decoder chooses its own representative M_0 from the visible advice. Every displayed advice map includes its affine intercept, through the coordinate t_0. The transferred slice is used only to analyze these private rules: the O-decoder does not use the U-decoder's witness or answer set. If the U-advice is outside $\mathcal{R}$, or if $\sigma|_H = 0$ for the O-decoder, that decoder returns an arbitrary valid answer.

Lemma 7.5 (A transferred slice gives agreement)**.** *Fix joint data* D *for which the chosen witness has the nonempty projected slice and agreement in Equation* (7.8)*. Conditional on* D*, the two decoders defined above, with independent randomness, satisfy* $\bar{\pi}a_U = a_O$ *with probability at least*

$$2^{-\ell r_{\mathrm{s}}}(\alpha/8)^2\, 2^{-r_{\mathrm{s}}}. \tag{7.10}$$

Proof. Put $Z' = \bar{\pi}Z$, so $\dim Z' \le r_{\mathrm{s}}$. The projected column specifications in the row fiber fix the restriction $N|_{Z'}$ to a linear map $b : Z' \to H$. The resulting affine slice is nonempty; choose one N_* in it. For every $\sigma \in K^*$ define

$$C_\sigma = \mathbb{E}\big[f_\sigma(N)(-1)^{\sigma(Nz')} \mid N|_{Z'} = b\big].$$

The character expansion of Equation (7.8) gives

$$\mathbb{E}_\sigma\big[(-1)^{\sigma(M_0 z'+u)}C_\sigma\big] \ge \alpha/4, \qquad \text{so} \qquad \mathbb{E}_\sigma|C_\sigma| \ge \alpha/4.$$

Exactly a fraction $2^{-\dim H}$ of the characters have trivial restriction to H. Since $\dim H \ge \ell - r_{\mathrm{s}}$ and $|C_\sigma| \le 1$, Equation (4.4) implies

$$\mathbb{E}_\sigma\big[\mathbf{1}_{\{\sigma|_H \neq 0\}}|C_\sigma|\big] \ge \alpha/8. \tag{7.11}$$

The annihilator in $\mathcal{N}^*$ of matrices vanishing on Z' is $H^* \otimes Z'$. Indeed, restriction $\mathcal{N} \to \mathrm{Hom}(Z', H)$ is onto, since a map on Z' extends after choosing a basis of L_O. Its dual image is precisely the functionals depending only on this restriction, generated by $N \mapsto \varphi(Nv)$ for $\varphi \in H^*$ and $v \in Z'$. Writing $Y = \{N \in \mathcal{N} : N|_{Z'} = 0\}$, the slice is $N_* + Y$. For any frequency $\Xi \in \mathcal{N}^*$, orthogonality on Y gives

$$\mathbb{E}_{N \in N_*+Y}(-1)^{\langle \Xi, N\rangle} = \mathbf{1}_{\{\Xi \in H^* \otimes Z'\}}(-1)^{\langle \Xi, N_*\rangle}.$$

For $\theta_\sigma = (\sigma|_H) \otimes z'$, Fourier expansion on the affine slice therefore gives

$$C_\sigma = \sum_{\Phi \in \theta_\sigma + H^* \otimes Z'} \widehat{f}_\sigma(\Phi)(-1)^{\langle \Phi + \theta_\sigma, N_*\rangle}.$$

This sum has $2^{\dim H \dim Z'} \le 2^{\ell r_{\mathrm{s}}}$ terms. Cauchy–Schwarz, first on each sum and then in the average over σ, shows from Equation (7.11) that

$$\begin{aligned}\mathbb{E}_\sigma \left[\mathbf{1}_{\{\sigma|_H \neq 0\}} \sum_{\Phi \in \theta_\sigma + H^* \otimes Z'} |\widehat{f}_\sigma(\Phi)|^2\right] \\ \ge 2^{-\ell r_{\mathrm{s}}}\mathbb{E}_\sigma\big[\mathbf{1}_{\{\sigma|_H \neq 0\}}C_\sigma^2\big] \ge 2^{-\ell r_{\mathrm{s}}}(\alpha/8)^2.\end{aligned}$$

This is a lower bound on the probability that the O-decoder samples a nontrivial restriction and a frequency in the indicated coset.

Every such frequency has the form $\Phi = (\sigma|_H) \otimes z' + \Psi$ with $\Psi \in H^* \otimes Z'$. For the decoder's choice of h it satisfies

$$\Phi(h) = z' + \Psi(h) \in z' + Z'.$$

As already proved, a frequency of positive mass gives first bit one. Equation (7.6) therefore places the returned vector in $\bar{\pi}\mathcal{A}$. Independently, the U-decoder samples uniformly from $\mathcal{A}$ and chooses a preimage with probability at least $1/|\mathcal{A}| \ge 2^{-r_{\mathrm{s}}}$. Multiplying the bounds proves Equation (7.10). □

Proof of Proposition 4.3. Use the two decoders above. Lemma 7.4 supplies a set of joint data of probability at least γ_1, and Lemma 7.5 gives the stated conditional agreement on that set. Thus one may take

$$\gamma = \gamma_1\, 2^{-\ell r_{\mathrm{s}}} (\alpha/8)^2\, 2^{-r_{\mathrm{s}}} > 0.$$

The largeness requirements for k come only from Theorem 4.1, the concentration bound in Lemma 7.2, and the total variation estimate in Lemma 7.3. They are uniform in the parity instance and table family. □

7.6 Choosing the tuple length

The threshold in Proposition 4.5 can be met by a terminating search over cubes once the map, noise, and shortcode data are fixed. The constants g_0, γ_1, and γ have been specified in Equation (7.3), Equation (7.7), and the proof just given. Write $m = 1 + 2k$ and $\beta = k^{-2/3}$; since k is a cube, β is rational. Require $m \geq m_0(\eta, \ell)$ and

$$\frac{\alpha^2}{32}\, 2^{(\ell - r_{\mathrm{s}})(m - r_{\mathrm{s}})} > (2r_{\mathrm{s}} + 1)(\ell + m).$$

The latter is a sufficient rational condition for the union bound in Lemma 7.2 to be less than one. For witness transfer in Lemma 7.4, require

$$\frac{1}{4}\left[\left(1 + \frac{|\mathcal{V}| - 1}{3}\beta^2\right)^k - 1\right] < \frac{\gamma_1^2}{(1 + \alpha/4)^2},$$

where γ_1 is from Equation (7.7). The left-hand side is the square of the total-variation bound. For the final contradiction, require

$$\left(1 - \frac{\beta\, 2^{-\dim W - r_{\mathrm{s}}}}{3600}\right)^k < \gamma,$$

with γ from Proposition 4.3. All these conditions are rational and hold for sufficiently large cubes k. Enumerating cubes until they hold therefore finds a suitable k; the search cost depends only on the two prescribed errors and is incurred before the input formula is used.

8 Singleton coordinates with zero advice slopes

Proposition 4.3 converts high acceptance of the matrix test into agreement between two local strategies supplied with correlated linear advice. We now bound the value of this advice experiment on a NO parity instance. The use of vanishing advice is part of the smooth outer-game framework of [26, Sections 3.3–3.4]. We give the full local reconstruction argument for the present occurrence-indexed questions. The key is to identify singleton coordinates on which both advice slopes vanish. After fixing the data outside those coordinates, their question pairs have the ordinary independent equation–variable law. Each side can then reconstruct its original advice input from its own remaining questions and the fixed data. This defines strategies for an ordinary parallel repetition, whose value bounds the original agreement.

We prove Lemma 4.4 for the experiment in Definition 4.2, with $k > 1$ an integer cube and $\beta = k^{-2/3}$. The distinct variable IDs in each equation make its ordinary equation–variable game $G_{\mathcal{E}}$ a projection game; Equation (3.1) gives its value at most $14/15$ when $\mathrm{OPT}(\mathcal{E}) \leq 4/5$.

Proof of Lemma 4.4. Fix an arbitrary pair of local randomized strategies in the advice experiment. We first condition on A and combine the two visible advice maps into

$$D_O = (T_O, S_O) : L_O \longrightarrow E_A, \qquad E_A = W \oplus \operatorname{im} A.$$

Conditional on the questions and projection choices, the intercept c_O and all block slope columns of D_O are independent uniform elements of E_A. Indeed, the columns of T_O and M_O are independent and uniform, and applying A to a uniform column gives a uniform element of $\operatorname{im} A$. This coefficient law depends only on whether a block is an equation or a singleton, and not on any equation ID, variable ID, or selected position. In particular, c_O is independent of all block data.

Let $I \subseteq [k]$ consist of the coordinates i projected to a singleton whose advice slope $d_i \in E_A$ is zero. Conditional on A, the indicators of membership in I are independent, with common probability

$$p_A = \frac{\beta}{|E_A|} = \beta\, 2^{-\dim W - \operatorname{rank} A}. \tag{8.1}$$

More precisely, for every equation occurrence e and position $j \in [3]$,

$$\Pr[e_i = e,\ i \text{ is a singleton at position } j,\ d_i = 0 \mid A] = \omega_e \frac{\beta}{3|E_A|}. \tag{8.2}$$

Thus, conditional on belonging to I, a coordinate has precisely the ordinary equation–variable distribution.

For the analysis, let R consist of A, I, the original intercept c_O, and the following data for every $i \notin I$: its equation occurrence, whether the block was retained or projected, the selected position in the latter case, and its advice slope columns. No witness, decoder output, agreement event, or response randomness is included in R.

For each value $R = r$ of positive probability, regard the set I recorded in r as fixed. When I is nonempty, we will use r as constant data when defining two strategies for $G_{\mathcal{E}}^{\otimes |I|}$. Each strategy will reconstruct one original advice input from its own product-game questions, apply the corresponding original response rule, and retain the answers in the slots I.

Conditional on $R = r$, the pairs

$$(e_i, v_{e_i, j_i}), \qquad i \in I,$$

are independent question pairs of $G_{\mathcal{E}}$. Indeed, conditional on A, the coordinate data are independent, and Equation (8.2) shows that conditioning on I preserves the ordinary law in each coordinate in I. The outside coordinate data and the independent intercept impose no further condition on these pairs.

We now verify the two reconstructions, including the affine intercept shared across the blocks. An equation with right-hand side b_e has actual bits $x, y, b_e t_0 + x + y$. Pulling back a singleton slope d_i thus produces the two slopes

$$(d_i, 0), \quad (0, d_i), \quad (d_i, d_i) \tag{8.3}$$

at positions one, two, and three, respectively. The intercept of $D_U = D_O \bar{\pi}$ is

$$c_U = c_O + \sum_{\substack{i \text{ projected to a singleton} \\ j_i = 3}} b_{e_i} d_i. \tag{8.4}$$

For a coordinate in I, every slope in Equation (8.3) is zero and its contribution to Equation (8.4) is zero. All remaining terms in that equation are determined by r.

The equation side receives its occurrences e_i for $i \in I$. Together with r, they specify all of U. Its two advice slopes in each block in I are zero, while r determines every other slope and the full intercept. These coefficients reconstruct both padded advice maps without using any selected position or variable question in I. The variable side receives the variable question for each $i \in I$. Together with r, these specify all of O. Its singleton slopes in I are zero, and r supplies every other slope and the original intercept. These coefficients reconstruct its two advice maps without using any equation in I. Repeated IDs across different blocks cause no additional restriction: questions were sampled independently with replacement, and answers use the local copies specified in Section 3.

These reconstructions define the promised product-game strategies: each side applies its original response rule to the reconstructed input and retains its answers in the slots I, in their original order. Their equation answers are satisfying triples because the original answers are valid. For every realization of the questions in I and response randomness, full agreement in the advice experiment implies acceptance in every slot in I. If $I = \varnothing$, the conditional success probability is at most one. Otherwise,

$$\Pr[\bar{\pi} a_U = a_O \mid R = r] \le \mathrm{val}(G_{\mathcal{E}}^{\otimes |I|}) \le \left(1 - \frac{1}{3600}\right)^{|I|}.$$

The first inequality uses the strategies just constructed and the conditional product law. The second uses Theorem 2.1 and Equation (3.1). The original response rules are evaluated only at their original local inputs; r indexes the auxiliary product-game strategies. Thus the final bound holds for every r, including $I = \varnothing$ when its right-hand side is interpreted as one.

Finally, conditional on A, the size of I has distribution $\mathrm{Binomial}(k, p_A)$. Averaging the preceding inequality first over R conditional on A, and then over A, gives

$$\begin{aligned}
\Pr\left[\bar{\pi}(a_U) = a_O\right] &\le \mathbb{E}_A \left(1 - \frac{p_A}{3600}\right)^k \\
&\le \left(1 - \frac{\beta\, 2^{-\dim W - r_{\mathrm{s}}}}{3600}\right)^k \\
&\le \exp\left(-\frac{k\beta\, 2^{-\dim W - r_{\mathrm{s}}}}{3600}\right),
\end{aligned}$$

since $\operatorname{rank} A \le r_{\mathrm{s}}$. Substituting $k\beta = k^{1/3}$ proves Equation (4.7). □

9 Approximation consequences

Theorem 1.1 turns established conditional hardness results into NP-hardness results. The underlying conditional reductions and approximation thresholds are due to the cited authors. We give the additional arguments needed for the exact instance models stated below. We also record two results from companion manuscripts. For a maximization problem, ratio r means returning value at least r times the optimum; for a minimization problem, factor c means returning value at most c times the optimum. All factors and tolerances below are fixed independently of input size. In each reduction using Theorem 1.1, they determine the Unique Games errors before its alphabet is chosen. If $\mathrm{P} \neq \mathrm{NP}$, the resulting hardness statements exclude deterministic polynomial-time approximations at the indicated strict thresholds.

9.1 Constraint satisfaction and Vertex Cover

For Max-Cut, which maximizes the number of edges crossing a bipartition, Khot, Kindler, Mossel, and O'Donnell's reduction gives NP-hardness of approximation at every fixed ratio in $(\alpha_{\mathrm{GW}}, 1)$,

where the Goemans–Williamson constant is $\alpha_{\mathrm{GW}} \simeq 0.87856$ [24, Theorem 1]. This matches the semidefinite approximation threshold of Goemans and Williamson [16]. The reduction uses the Majority Is Stablest theorem of Mossel, O'Donnell, and Oleszkiewicz [32]. For Vertex Cover, which minimizes the number of vertices meeting every edge, Khot and Regev's reduction gives NP-hardness at every fixed factor in $(1, 2)$ [27].

A constraint satisfaction problem (CSP) is specified here by a finite nonempty domain D and a finite family Λ of Boolean predicates on D, of bounded arity. A weighted Λ-instance has a finite variable set and a finite collection of predicates from Λ applied to tuples of variables, with nonnegative rational weights summing to one. An assignment maps the variables to D; its value is the total satisfied weight, and OPT is the maximum of this value over all assignments. Raghavendra's theorem characterizes the optimal approximation for every fixed language of this form, up to arbitrarily small positive error, by a basic semidefinite relaxation [36, Theorem 1.1 and Corollary 1.3]. Write LC for the value of Raghavendra's LC relaxation, as defined in [37, Section 4.5].

The fixed-instance CSP theorem and the ordering reduction use the following vertex-wise completeness guarantee.

Orient the constraints of a bipartite game from its left side to its right side. A left vertex is *strongly satisfied* by a labeling when every edge incident to it is satisfied.

Lemma 9.1 (Simple games with completeness on left vertices)**.** *Fix $0 < \zeta, \nu < 1$ and integers $d, H \geq 1$. There are integers $h, R \geq 1$, with $h \geq H$ and $d \mid h$, and a deterministic polynomial-time reduction from* 3SAT *to nonempty explicit unweighted simple bipartite Unique Games instances $\Phi_\varphi = (A, B, E)$ over $[R]$ with the following properties. Every left vertex has degree h. On a satisfiable input φ, one labeling strongly satisfies at least $(1 - \zeta)|A|$ left vertices. On an unsatisfiable input, $\mathrm{val}(\Phi_\varphi) < \nu$. The degree h and the two errors used in Theorem 1.1 may be chosen from ζ, ν, d, H before that theorem supplies the alphabet R.*

Appendix A.1 proves this formulation, with the errors and left degree chosen before the alphabet.

Corollary 9.2 (Raghavendra's consequence)**.** *Fix a finite domain D, a finite bounded-arity Boolean predicate family Λ on D, and a weighted Λ-instance J in the rational model above. Suppose $A = \mathrm{OPT}(J) < B = \mathrm{LC}(J)$. For every fixed $0 < \eta < (B - A)/2$, it is NP-hard to distinguish Λ-instances I satisfying*

$$\mathrm{OPT}(I) \geq B - \eta \qquad \textit{from those satisfying} \qquad \mathrm{OPT}(I) \leq A + \eta.$$

The proof in Appendix A.2 applies the fixed-instance theorem with tolerance $\eta/4$ and changes its fixed finite verifier law by total variation less than $\eta/2$. The latter change yields rational weights and changes every assignment value by less than $\eta/2$.

9.2 Ordering constraints

An ordering CSP uses a global order of the input variables, rather than an assignment from a fixed finite domain. Fix an integer $k \geq 2$ and a nonempty proper subset $\Pi \subset S_k$, where S_k is the set of permutations of $\{1, \ldots, k\}$. In the finite probability-weighted model of Guruswami, Håstad, Manokaran, Raghavendra, and Charikar [17, Definition 8.2], we use an explicitly listed finite variable set V and a finite list of constraint occurrences. Their nonnegative rational weights are encoded by binary numerators and denominators and sum to one. Every occurrence is an ordered k-tuple $(v_1, \ldots, v_k)$ of distinct variables, satisfied by a linear order $\prec$ when

$$v_{\pi(1)} \prec \cdots \prec v_{\pi(k)} \quad \text{for some } \pi \in \Pi.$$

Write $\mathrm{OPT}_\Pi(I)$ for the maximum satisfied weight. A uniform random order satisfies expected weight $\rho_\Pi = |\Pi|/k!$. Since $\mathrm{OPT}_\Pi(I) \le 1$, it is a ρ_Π-approximation. For fixed k, conditional expectation gives a deterministic polynomial-time algorithm with this guarantee. The end of Appendix A.3 gives the exact conditional probabilities and the bit-length bound. The size of V is part of the input, so this is not a CSP over the fixed finite domain used in the preceding corollary.

Proposition 9.3 (Finite ordering transfer)**.** *Fix $k \ge 2$ and $\varnothing \ne \Pi \subsetneq S_k$, put $\rho_\Pi = |\Pi|/k!$, and fix $0 < \delta < (1-\rho_\Pi)/2$. There are integers $q, H \ge 1$ and positive rationals $\zeta, \nu < 1$, depending only on k, Π, δ, with the following property for every fixed integer $R \ge 1$.*

Let $\Phi = (A, B, E)$ be a nonempty explicit unweighted simple bipartite Unique Games instance over $[R]$, oriented from A to B, whose common left degree h satisfies $h \ge H$ and $q \mid h$. There is a deterministic transformation, polynomial in the explicit size of Φ for fixed R, producing an instance I given by an explicit list of Π-ordering constraints on distinct variables within each constraint, with nonnegative rational weights, each encoded by a binary numerator and a positive binary denominator, summing to one. If some labeling strongly satisfies at least $(1-\zeta)|A|$ left vertices, then $\mathrm{OPT}_\Pi(I) \ge 1-\delta$. If $\mathrm{val}(\Phi) \le \nu$, then $\mathrm{OPT}_\Pi(I) \le \rho_\Pi + \delta$. The constants q, H, ζ, ν are independent of both R and h.

Appendix A.3 proves the proposition. The degree conditions can be supplied by Lemma 9.1 before its alphabet is chosen.

Corollary 9.4 (Ordering-CSP consequence)**.** *Fix $k \ge 2$ and $\varnothing \ne \Pi \subsetneq S_k$, and put $\rho_\Pi = |\Pi|/k!$. For every fixed $0 < \varepsilon < (1-\rho_\Pi)/2$, it is NP-hard to distinguish finite weighted Π-ordering-CSP instances I satisfying*

$$\mathrm{OPT}_\Pi(I) \ge 1-\varepsilon \qquad \textit{from those satisfying} \qquad \mathrm{OPT}_\Pi(I) \le \rho_\Pi + \varepsilon.$$

Consequently, approximation at every fixed ratio $r \in (\rho_\Pi, 1]$ is NP-hard. Together with the random-ordering guarantee, this gives the optimal threshold ρ_Π.

Proof. Apply Proposition 9.3 with $\delta = \varepsilon$. It uses the local gap and analytic estimates of Guruswami et al. [17] and supplies q, H, ζ, ν, depending only on k, Π, ε, before any alphabet is chosen. Apply Lemma 9.1 with these ζ, ν, with $d = q$, and with the same H. The degree and errors are fixed before the alphabet is chosen. Using Theorem 1.1, it maps each 3SAT formula φ to a simple game with completeness on left vertices $\Phi_\varphi = (A, B, E)$ over a fixed alphabet $[R]$ whose common left degree h satisfies $h \ge H$ and $q \mid h$. If φ is satisfiable, one labeling strongly satisfies at least $(1-\zeta)|A|$ left vertices; otherwise $\mathrm{val}(\Phi_\varphi) < \nu$. These are the hypotheses of Proposition 9.3 for this fixed R, so composing the two reductions gives $\mathrm{OPT}_\Pi(I) \ge 1-\varepsilon$ in the first case and $\mathrm{OPT}_\Pi(I) \le \rho_\Pi + \varepsilon$ in the second. This proves the displayed gap for the original predicate Π.

For a fixed $r \in (\rho_\Pi, 1]$, choose the positive error still smaller so that $\varepsilon < (r - \rho_\Pi)/(1+r)$. Then $r(1-\varepsilon) > \rho_\Pi + \varepsilon$, so an r-approximation would distinguish the two cases. □

For Maximum Acyclic Subgraph in the finite weighted directed-graph model, take $k = 2$ and $\Pi = \{12\}$ in one-line notation. A constraint (u, v) asks that the edge point forward; forward edges form an acyclic subgraph, and every acyclic subgraph has a topological order. Its threshold is therefore $1/2$. For Betweenness, the constraint (a, b, c) asks that b lie between a and c, so $\Pi = \{123, 321\} \subset S_3$ and the threshold is $2/3! = 1/3$ [17, Theorem 1.1 and Section 1.2]. These finite weighted consequences have near-perfect completeness with fixed positive error.

9.3 Cut, deletion, and clustering consequences

For the cut problems of Chawla, Krauthgamer, Kumar, Rabani, and Sivakumar [9, Section 1], let $G = (V, E)$ be a finite undirected graph with positive rational edge costs c_e, and let $\mathcal{D} = \{\{s_i, t_i\} : 1 \le i \le k\}$ contain $k \ge 1$ distinct demand pairs with $s_i \ne t_i$. For $M \subseteq E$, write $c(M) = \sum_{e \in M} c_e$, and let $q_{\mathcal{D}}(M)$ count the demand pairs disconnected in $G - M$. Weighted Multicut minimizes $c(M)$ subject to $q_{\mathcal{D}}(M) = k$. Their nonuniform Sparsest Cut search problem minimizes $c(M)/q_{\mathcal{D}}(M)$ over cutsets M with $q_{\mathcal{D}}(M) > 0$. Only the selected pairs carry unit demand; they need not be all vertex pairs.

For weighted Min-2CNF$^{\equiv}$ Deletion, the input consists of clauses $(\ell \equiv \ell')$ with positive rational weights, where ℓ, ℓ' are Boolean literals, and an assignment pays the total weight of false clauses. The allowed clauses therefore express equality or disequality of variables.

In the arbitrary weighted-graph form of Correlation Clustering, each present edge has a $+$ or $-$ label and a nonnegative rational weight. A partition into any number of clusters pays for $+$ edges whose endpoints lie in different clusters and $-$ edges whose endpoints lie in the same cluster. Missing edges carry no penalty. This is the minimum-disagreements objective of [10, Section 2.1].

Corollary 9.5 (Cut, deletion, and clustering hardness)**.** *For every fixed real constant $C > 1$, it is NP-hard to approximate weighted* Multicut*, nonuniform* Sparsest Cut*, weighted Min-*2CNF$^{\equiv}$ *Deletion, and weighted* Correlation Clustering*, in the models above, within factor C. For* Sparsest Cut*, this is hardness of the search task under polynomial-time Cook reductions: the algorithm must output a cutset, not only an estimate of the optimum.*

Proof. Corollary 1.3 of [9] gives the first three conclusions under its ordinary edge-value Unique Games hypothesis. For a requested factor C, use its hardness statement at any fixed larger factor L: a C-approximation would also be an L-approximation. To supply that hypothesis, orient the bipartite constraints of Theorem 1.1 left-to-right, renumber its fixed alphabet as $[d]$ with $d = 2^s \ge 2$, and give its edges equal weights summing to one. The unique 2-prover games of [9, Section 1.2] have the same number of questions on each side, so pad the smaller vertex side with isolated questions if needed; this preserves both value and polynomial size. The search-version qualification for Sparsest Cut is in Section 1.3 of the cited source.

For Correlation Clustering, the hard Multicut instances in [9, Section 2.1] have distinct antipodal demand pairs inside d-cubes, while the edges of the Multicut graph are cube edges or join different cubes. Lemma 1.5 of that source, which makes the game regular, does not change d, so $d \ge 2$ there. Antipodal vertices of a d-cube differ in all $d \ge 2$ coordinates, whereas a cube edge joins vertices that differ in exactly one; edges between cubes cannot join a same-cube demand pair either. Thus the demand pairs are nonedges. The transformation of Demaine, Emanuel, Fiat, and Immorlica [10, Theorem 4.7 and Corollary 4.8] therefore creates no opposite-signed parallel edges. It preserves the optimum and converts every clustering into a multicut of no greater cost, so the same fixed-factor hardness transfers. To see the conversion explicitly, put $W_+ = \sum_{e \in E} c_e$. Keep the original edges as positive edges with their costs, and give each demand pair a negative edge of weight $W_+ + 1$. If a clustering separates every demand pair, its crossing positive edges form a multicut of exactly its disagreement cost. If it leaves a demand pair together, its cost exceeds W_+; returning all original edges is then a multicut of smaller cost. Conversely, the connected components left by a multicut form a clustering of cost at most that multicut's cost. Taking minima in both directions proves equality of optima. The sums and the added weights have polynomial binary length, so this also preserves the rational input model. □

The separate manuscript [33, Theorem 1.1] proves NP-hardness at every fixed factor $C > 1$ for

the bipartition objective with nonnegative rational capacities and exactly unit demand between every pair of distinct vertices.

Kernel clustering. A separate companion application concerns the following maximization problem. For *identity-target kernel clustering*, fix an integer $k \geq 3$. An input is an explicitly represented rational symmetric matrix $A = (a_{pq})_{p,q=1}^{N} \succeq 0$ with $A\mathbf{1} = 0$, where $\mathbf{1}$ is the all-ones vector. An assignment $\sigma : \{1, \ldots, N\} \to \{1, \ldots, k\}$ has value

$$\mathrm{val}_A(\sigma) = \sum_{p,q=1}^{N} a_{pq} \mathbf{1}_{\{\sigma(p)=\sigma(q)\}}, \qquad \mathrm{OPT}_k(A) = \max_{\sigma} \mathrm{val}_A(\sigma).$$

The sum includes ordered pairs and the diagonal; empty clusters are allowed, and there is no normalization by cluster size. A loss factor $\alpha \geq 1$ means returning value at least $\mathrm{OPT}_k(A)/\alpha$. The Gaussian propeller companion uses Theorem 1.1 to prove NP-hardness of achieving every fixed loss factor

$$1 \leq \alpha < \alpha_k, \qquad \alpha_k = \frac{8\pi}{9}\left(1 - \frac{1}{k}\right),$$

for each fixed $k \geq 3$ [35, Theorem 6.1]. This is strict hardness below the rounding factor α_k.

A Finite formulations of the consequences

Some of the reductions in Section 9 use a bipartite Unique Games instance in which a nearly satisfying labeling satisfies all constraints incident to most left vertices. We first obtain a simple graph with this property and with a prescribed lower bound on its common left degree. We use this formulation for the rational CSP conversion and the finite ordering reduction. The local ordering construction follows Guruswami, Håstad, Manokaran, Raghavendra, and Charikar [17] and uses the low-influence results cited below. The composition specifies the graph, the independent neighbor choices, and the conversion to constraints on distinct variables.

A.1 Simple Unique Games with completeness on left vertices

Proof of Lemma 9.1. We adapt the normalization and sequence construction of Khot and Regev [27, Lemmas 3.3, 3.4, and 3.6]. The extra copies of right vertices used below ensure that the final graph is simple.

Choose a multiple h of d with

$$h \geq \max\{H, 2\}, \qquad \frac{1}{h} < \frac{\nu}{4}.$$

For $(T)_h = T(T-1)\cdots(T-h+1)$, choose a fixed integer $T \geq h$ such that

$$p_{\mathrm{graph}} := \frac{(T)_h}{T^h} > \frac{3}{4}.$$

Such a choice exists because $1 - (T)_h/T^h \leq \binom{h}{2}/T$. Choose positive rationals β, γ, u and a positive integer L_2 so that

$$\beta < \frac{\nu}{4}, \qquad h^2\gamma < \frac{\nu}{4}, \qquad (h+1)u < \frac{\zeta}{2}, \qquad \frac{1}{L_2} < \min\left\{\frac{\beta\gamma}{2}, \frac{\zeta}{2h}\right\}.$$

Finally choose positive rational $\gamma_0, \zeta_0 < 1/2$ with

$$2\gamma_0 < \frac{\beta\gamma}{2}, \qquad \sqrt{2\zeta_0} < u.$$

All these choices precede the invocation of Theorem 1.1 with completeness error ζ_0 and soundness error γ_0.

Write its output as $\Phi_0 = (X_0, Y, E_0)$, orienting each permutation left-to-right and renumbering the fixed alphabet as $[R]$. Put $M_0 = |E_0| > 0$, give each edge weight $1/M_0$, and let w_x be the total incident weight at x. Simplicity gives one permutation at each pair of positive weight. Create

$$n_x = \lfloor 2|X_0|w_x \rfloor$$

copies of x, each with local edge weights w_{xy}/w_x. Only vertices with $w_x > 0$ have copies. If X_1 is the copy set, then

$$|X_0| \le |X_1| = \sum_x n_x \le 2|X_0|,$$

and the incident weight at each copy is one.

Here and below a local satisfied weight uses the labels at that left vertex and at the shared right vertices. Given any labeling of the copy game, keep the right labels and give all copies of each x a label that maximizes their common local objective. The corresponding labeling L of Φ_0 satisfies

$$\sum_{x' \in X_1} w_{\mathrm{sat}}(x') \le \sum_{x : n_x > 0} \frac{n_x}{w_x} w_{\mathrm{sat},L}(x) \le 2|X_0|\, w_{\mathrm{sat},L}(\Phi_0).$$

Consequently the copy game's normalized value is at most $2\gamma_0$ on a NO input. On a YES input, lift a labeling of Φ_0 with unsatisfied weight at most ζ_0. The same calculation for unsatisfied weight gives mean local loss at most $2\zeta_0 < u^2$. Markov's inequality leaves more than a $1 - u$ fraction of X_1 with local satisfied weight at least $1 - u$.

To remove the weights, put $\alpha = L_2|Y|$. At each $x \in X_1$, write $\widetilde{w}_{xy}$ for its normalized local weights. Choose one neighbor y_0 of positive local weight, round $\alpha\widetilde{w}_{xy}$ down for every $y \ne y_0$, and assign the remaining occurrences to y_0. All occurrences at a pair retain that pair's permutation. This gives an unweighted multigraph Φ_{occ} of left degree α. For any subset of the neighbors, the difference between its old weight and its new occurrence fraction has magnitude at most $|Y|/\alpha = 1/L_2$: sum the rounding losses over the subset if it omits y_0, and over its complement otherwise. This applies to the satisfied subset for every labeling. Hence

$$\mathrm{val}(\Phi_{\mathrm{occ}}) \le 2\gamma_0 + \frac{1}{L_2} < \beta\gamma \quad \text{on a NO input.} \tag{A.1}$$

On a YES input, the labeling above has satisfied occurrence fraction at least $1-a$, where $a = u+1/L_2$, at more than a $1 - u$ fraction of X_1.

Index the α occurrences at x by j, with right endpoint y_j and permutation π_j. Replace each $y \in Y$ by (y, t), $t \in [T]$, and replace occurrence j by all T occurrences from x to (y_j, t) with permutation π_j. Call the resulting left-αT-regular multigraph Φ_{tag}. For any labeling of Φ_{tag}, keep its left labels and, independently for each original y, give y the label of a uniformly chosen copy (y, t). The expected edge value in Φ_{occ}, counting its occurrences, is exactly the given edge value in Φ_{tag}. Thus

$$\mathrm{val}(\Phi_{\mathrm{tag}}) \le \mathrm{val}(\Phi_{\mathrm{occ}}) < \beta\gamma \quad \text{on a NO input.} \tag{A.2}$$

The tag-independent lift of the YES labeling preserves every local satisfied fraction.

Consider the full sequence graph whose left vertices are

$$(x, (j_1, t_1), \ldots, (j_h, t_h)), \qquad j_i \in [\alpha], \quad t_i \in [T],$$

with position i joined to (y_{j_i}, t_i) by π_{j_i}. This is the sequence construction of Khot and Regev; its left vertex identifier includes the entire ordered occurrence sequence. We record the soundness estimate in a form that permits the left label to depend on that entire sequence. Each x contributes exactly $(\alpha T)^h$ full sequence vertices, all of degree h. Thus the uniform edge value is the average satisfied fraction over a uniform x, an ordered h-tuple of independent uniform occurrences at x, and a uniform position in that tuple.

Fix arbitrary right labels in Φ_{tag}. Each occurrence at x determines the unique left label that satisfies it. Let $p_x(\ell)$ be the fraction determining label ℓ. The $p_x(\ell)$ sum to one, and choosing the best left label at each x gives

$$\mathbb{E}_x \max_{\ell} p_x(\ell) \le \mathrm{val}(\Phi_{\mathrm{tag}}) < \beta\gamma.$$

Fewer than a β fraction of x therefore have $\max_\ell p_x(\ell) \ge \gamma$. At every other x, the probability that two labels determined by h independent occurrences agree is at most

$$\binom{h}{2} \sum_{\ell} p_x(\ell)^2 \le \binom{h}{2} \max_{\ell} p_x(\ell) < h^2\gamma.$$

If no two determined labels agree, any label of the sequence vertex satisfies at most one of its h edges. On the remaining sequences its satisfied fraction is at most one. Averaging first over sequences and then over x, and using $1/h + h^2\gamma < 1$, bounds the full sequence graph's value by

$$s := \beta + \frac{1}{h} + (1-\beta)h^2\gamma < \frac{3\nu}{4}. \tag{A.3}$$

Retain only sequence vertices for which $t_1, \ldots, t_h$ are pairwise distinct. For each x and each base sequence $(j_1, \ldots, j_h)$, exactly $(T)_h$ tag sequences remain. Each x thus contributes $\alpha^h(T)_h$ retained vertices, a fraction p_{graph} of its full set. The h right neighbors of each retained vertex are distinct because their tags are distinct. Different occurrence sequences remain different left vertices, so this retained graph is simple and has common left degree h.

Give each sequence vertex based at x the old label of x, and keep the tag-independent right labels. For this lifted YES labeling, satisfaction at position i depends only on j_i. Conditioning the tags to be distinct leaves the base occurrence sequence independent and uniform. If the satisfied fraction at x in Φ_{occ} is f_x, exactly an f_x^h fraction of its retained sequence vertices are strongly satisfied. Since every x contributes equally, the strongly satisfied fraction is at least

$$(1-u)(1-a)^h \ge 1 - u - ha = 1 - (h+1)u - \frac{h}{L_2} > 1 - \zeta.$$

Here $0 < a < 1$ follows from the parameter choices.

For NO, extend any labeling of the retained graph to the full graph by giving discarded left vertices arbitrary labels and keeping all right labels. All left degrees are h, so its full edge value is at least p_{graph} times its retained edge value. By (A.3), the latter is at most

$$\frac{s}{p_{\mathrm{graph}}} < \frac{3\nu/4}{3/4} = \nu.$$

All the operations are finite and deterministic. If N bounds the explicit size of Φ_0, then $|X_1| = O(N)$ and $\alpha = L_2|Y| = O(N)$. The retained graph has $|X_1|\alpha^h(T)_h = O(N^{h+1})$ left vertices and h edges per left vertex. Its identifiers and the rational arithmetic used for rounding have polynomial bit length, and its permutation tables use the fixed alphabet inherited from Φ_0. This proves the claimed polynomial bound and completes the reduction. □

A.2 Rational weights for the CSP consequence

Proof of Corollary 9.2. For the thesis's common-arity convention [37, Definitions 7.2.1–7.2.2], identify D with a fixed $[q]$, and pad every predicate with ignored trailing arguments to a common positive arity, filling the added positions with a fixed variable of J. Such a variable exists because $A < B$: an instance without variables applies only constant predicates, so its optimum and its relaxation value would both equal its total weight of true predicates. This leaves each applied payoff and its dependence set unchanged, hence preserves OPT and the relaxation of [37, Section 4.5]. In that relaxation, each local distribution is over assignments to the variables on which its applied payoff actually depends. When emitting an occurrence, discard only the added trailing positions, retaining the original predicate from Λ and its weight.

Apply Raghavendra's fixed-instance theorem [37, Theorem 7.9] with tolerance $\eta/4$; see also [36, Theorem 1.1 and Lemma 2.3]. That theorem asserts UG-hardness in the sense of [37, Definition 2.5.4]. Its reduction, the composition of Theorems 7.5 and 7.6 there, starts from [37, Problem 9]: bipartite Unique Games instances in which one labeling satisfies every edge at a $(1-\gamma)$ fraction of the left vertices are to be distinguished from instances in which no labeling satisfies more than a δ_{U} fraction of the edges. The bounds of [37, Theorem 7.6] give the gap below once $\gamma, \delta_{\mathrm{U}} > 0$ are small enough in terms of J and η; fix such errors. Lemma 9.1, with $\zeta = \gamma$, $\nu = \delta_{\mathrm{U}}$ and $d = H = 1$, and Theorem 1.1 supply exactly these instances, with a fixed alphabet $[M]$; strong satisfaction of a left vertex there is the notion of [37, Definition 2.5.3]. Every left vertex has the same degree, so the thesis's outer verifier [37, Section 7.5], which picks a uniform left vertex and then a uniform neighbor, samples a uniform edge, and its edge value is the value val used here. Its verifier instances have optimum at least $B - \eta/4 > B - \eta/2$ in the YES case and at most $A + \eta/4 \le A + \eta/2$ in the NO case.

We pass from the verifier's probability weights to the stated rational weights. For this fixed invocation, the construction in [37, Sections 7.3.1 and 7.5] gives a local probability law θ on a finite set $\mathcal{S}$, after the alphabet $[M]$ is fixed. A record $s \in \mathcal{S}$ specifies an applied predicate of J, its $a(s)$ ordered oracle-query strings in D^M, and the positions filled by their replies. There is one query for each distinct variable on which the applied payoff depends. If that variable occupies several predicate positions, the same reply fills all those positions, just as in the original applied predicate. Thus independent neighbor sampling is per query, not per syntactic occurrence of a repeated variable. This law may depend on J, η, M, but no varying input parameter enters it. Rational probability vectors are dense in this finite simplex, so fix $\widehat{\theta} \in \mathbb{Q}^{\mathcal{S}}_{\ge 0}$ with

$$\sum_{s\in\mathcal{S}} \widehat{\theta}(s) = 1, \qquad \delta := \frac{1}{2}\sum_{s\in\mathcal{S}} |\widehat{\theta}(s) - \theta(s)| < \frac{\eta}{2}.$$

This rational table is fixed finite data of the reduction; its description length may depend on J, η, M.

Let L_Φ and V_Φ be the left and right sides of the nonempty simple game supplied above, and let $h \ge 1$ be its common left degree. List the variables $V_\Phi \times D^M$ used in Section 7.5 of the cited thesis. Its outer verifier chooses a uniform left vertex u and, independently for each oracle query, chooses a uniform neighbor of u and relabels the query string by that edge's permutation. For each $s \in \mathcal{S}$, $u \in L_\Phi$, and ordered neighbor tuple in $N(u)^{a(s)}$, emit the Λ-predicate prescribed by s on the resulting query variables. Fill all positions of any syntactic variable omitted from the test with the first listed output variable; the applied predicate is independent of such variables, so this leaves its value unchanged. Give the occurrence weight

$$\frac{\widehat{\theta}(s)}{|L_\Phi| h^{a(s)}}.$$

These weights are nonnegative rationals summing to one. Keeping equal occurrences separate gives $|L_\Phi| \sum_{s\in\mathcal{S}} h^{a(s)}$ occurrences and $|V_\Phi||D|^M$ variables. The local set and every $a(s)$ are fixed, while the graph has polynomial size; hence this is a polynomial occurrence list. The explicit neighbor lists and edge permutations let us write it in polynomial time. Each weight has polynomial binary length: it can be represented with a fixed denominator multiplied by $|L_\Phi|h^{a(s)}$.

For an assignment σ, let $g_\sigma(s) \in [0,1]$ be its conditional acceptance probability over these outer choices. The original and rationalized instances use the same variable set, and

$$|\mathrm{val}_{\widehat{\theta}}(\sigma) - \mathrm{val}_\theta(\sigma)| = \left|\sum_{s\in\mathcal{S}} (\widehat{\theta}(s) - \theta(s)) g_\sigma(s)\right| \le \delta < \frac{\eta}{2}.$$

Taking maxima gives the same bound for their optima. The imported gap therefore yields optimum at least $B - \eta$ in the YES case and at most $A + \eta$ in the NO case, as required. □

A.3 The finite ordering transfer

The ordering theorem of Guruswami et al. uses a multiscale local instance and a low-influence rounding estimate. Their general outer composition is described as analogous to the two-query construction in their Section 7. Proposition 9.3 gives the composition for the rational occurrence-list model of Section 9. It separates the constants chosen for the ordering gap from the subsequently chosen Unique Games alphabet.

The output variables will be $(b, z) \in B \times [m]^R$, for a fixed local symbol set $[m]$ chosen below. At each $a \in A$, restrict a global order to $N(a) \times [m]^R$ and divide that local sorted list into q equal consecutive blocks. Averaging block membership over neighbors, with their edge permutations acting on z, gives one simplex-valued function of z, whose components have mean $1/q$ under uniform z. For fixed a, conditional on the local query strings, independent neighbor choices make the block ranks independent with probabilities given by this function. The local estimate bounds their expected coarse payoff under small influences; the collision estimate allows correlated strings and bounds the chance of a shared block, where coarsening can change the induced order. A large influence in an averaged rank interval instead yields a game label, and many such vertices contradict low edge value. Lemma A.1 supplies the local gap and common Gram data, Lemma A.3 bounds the coarse payoff, and Lemma A.2 bounds the loss from equal coarse ranks. The proof of Proposition 9.3 combines these in Equation (A.23), after conditioning on distinct neighbors and refining the coarse completeness map to a strict order.

For $P = \mathbf{1}_\Pi$, choose $\pi_0 \in \Pi$ and set

$$P_{\mathrm{id}}(\sigma) = P(\pi_0 \circ \sigma), \qquad \sigma \in S_k.$$

Then $P_{\mathrm{id}}(\mathrm{id}) = 1$, and its average is ρ_Π. For a vector of integer ranks $r = (r_1, \dots, r_k)$, let $\mathrm{Ref}(r)$ be the permutations compatible with their weak order: they list positions in increasing rank and may order positions of equal rank arbitrarily. Define

$$\overline{P_{\mathrm{id}}}(r) = \frac{1}{|\mathrm{Ref}(r)|} \sum_{\sigma\in\mathrm{Ref}(r)} P_{\mathrm{id}}(\sigma). \tag{A.4}$$

This is the uniform tie extension used in [17, Section 8.2]. It lies in $[0,1]$ and agrees with P_{id} when the ranks are distinct.

The normalization preserves the predicate after one permutation of tuple positions. Given a raw tuple $w = (w_1, \dots, w_k)$, define the emitted tuple v by

$$v_{\pi_0(j)} = w_j \qquad (j \in [k]). \tag{A.5}$$

If a strict order induces σ on w, it induces $\pi_0 \circ \sigma$ on v. More generally, relabeling positions gives the bijection

$$\mathrm{Ref}(f(v_1), \dots, f(v_k)) = \{\pi_0 \circ \sigma : \sigma \in \mathrm{Ref}(f(w_1), \dots, f(w_k))\}$$

for every rank map f. Uniform averages over these sets therefore give equal P- and P_{id}-payoffs, including ties. The permutation of positions in (A.5) preserves distinctness and weights.

For a finite local instance $J = (U, \mathcal{C}, \omega)$, write $t_c = (u_{c,1}, \dots, u_{c,k})$ for the tuple of occurrence $c \in \mathcal{C}$ and ω_c for its probability weight. Its q-ordering value is

$$\mathrm{opt}_q(J) = \max_{g:U\to[q]} \sum_{c\in\mathcal{C}} \omega_c \, \overline{P_{\mathrm{id}}}(g(u_{c,1}), \dots, g(u_{c,k})).$$

Lemma A.1 (Rational local ordering data)**.** *For every integer $q \geq 1$ and rational $\eta > 0$, there are a finite local instance $J = (U, \mathcal{C}, \omega)$, with $|U| = m \geq 2$, and probability laws μ_c on $[m]^k$ with the following properties. Every occurrence $c \in \mathcal{C}$ has a tuple t_c of k distinct variables; the weights ω_c and all atoms of μ_c are rational. Every one-position marginal of μ_c is uniform on $[m]$, and there is a fixed $0 < \alpha \leq 1/2$ such that $\mu_c(x) > \alpha$ for every c, x. Moreover*

$$\mathrm{opt}_q(J) \leq \rho_\Pi + \eta, \qquad \sum_c \omega_c \mathbb{E}_{X\sim\mu_c} \overline{P_{\mathrm{id}}}(X) \geq 1 - \eta. \tag{A.6}$$

The local laws have common feasible Gram data for the relaxation in [17, Section 8.3]: there are a unit vector I and vectors $b_{u,j}$, $u \in U, j \in [m]$, with

$$\sum_j b_{u,j} = I, \quad \langle b_{u,i}, b_{u,j}\rangle = 0 \ (i \neq j), \quad \|b_{u,j}\|^2 = \frac{1}{m},$$

and, for every two positions r, s of c,

$$\langle b_{u_{c,r},i}, b_{u_{c,s},j}\rangle = \Pr_{X\sim\mu_c}[X_r = i, \ X_s = j]. \tag{A.7}$$

All these data are fixed by k, Π, q, η.

Proof. The explicit increasing-tuple construction in [17, Theorem 11.1, Definition 11.2, and the proof of Theorem 2.5], applied to P_{id}, gives a finite instance J^* with $\mathrm{opt}_q(J^*) \leq \rho_\Pi + \eta$ and a linear order satisfying every constraint. More explicitly, use Theorem 11.1 with error $\eta/k!$. For every q-valued rank map, each of the $k!$ permutations under uniform tie refinement has probability at most $1/k! + \eta/k!$. Summing over the $|\Pi|$ permutations accepted by P_{id} gives at most $\rho_\Pi + \eta$. All raw tuples are increasing and $P_{\mathrm{id}}(\mathrm{id}) = 1$, so their natural order satisfies every constraint. Its distribution uses only uniform choices from finite sets, so its weights are rational. Its tuple positions lie in successive disjoint intervals and are therefore distinct.

Use the finite-copy smoothing from the proof of [17, Lemma 8.9], specialized to this complete order. Take L disjoint copies of J^*, each of weight $1/L$, and place their complete orders consecutively in an order $\sigma : U \to [m]$, where $m = L|U^*|$. The resulting instance J still has $\mathrm{opt}_q(J) \leq \rho_\Pi + \eta$, since the value of any q-assignment is the average of its values on the copies. Choose L and a positive rational $s_0 < 1$ so that $1/L + s_0 < \eta$. Define a random assignment $X : U \to [m]$ as follows: with probability $1 - s_0$, choose a uniform cyclic shift of the ranks of σ; with probability s_0, assign the variables independently and uniformly from $[m]$.

Each variable is uniform in either part of the mixture. A cyclic cut can split at most one of the L consecutive copies, so every shift has payoff at least $1 - 1/L$. The mixture's payoff is therefore at

least $(1-s_0)(1-1/L) > 1-\eta$. Let μ_c be its restriction to the variables of c. If $n_c(x)$ is the number of the m cyclic shifts with restriction x, then

$$\mu_c(x) = (1-s_0)\frac{n_c(x)}{m} + \frac{s_0}{m^k}.$$

These atoms are rational and exceed $\alpha := s_0/(2m^k)$. The denominator m^k is valid because the variables in c are distinct.

Finally regard the finite mixture as a probability space and take $I = \mathbf{1}$ and $b_{u,j} = \mathbf{1}_{\{X(u)=j\}}$ in its L^2 space. Uniform marginals, disjoint events for different labels of one variable, and restrictions to each c give all the displayed Gram identities. This constructs the required feasible data for this particular local instance. □

For scalar functions on the uniform product $[m]^R$, use

$$\mathrm{Inf}_\ell(F) = \mathbb{E}[\mathrm{Var}_{z_\ell}(F \mid z_{-\ell})]\,.$$

The noise operator T_θ, $0 \le \theta \le 1$, independently keeps each coordinate with probability θ and otherwise resamples it uniformly, then averages the function. In an orthonormal product expansion these are

$$T_\theta F = \sum_\sigma \theta^{|\sigma|}\widehat{F}(\sigma)\chi_\sigma, \qquad \mathrm{Inf}_\ell(F) = \sum_{\sigma:\sigma_\ell\neq 0} \widehat{F}(\sigma)^2. \tag{A.8}$$

Here χ_σ ranges over a fixed orthonormal product basis for uniform measure on $[m]^R$, indexed by $\sigma \in \{0,\dots,m-1\}^R$. Index zero denotes the constant basis element in each coordinate, $|\sigma|$ counts the nonzero indices, and $\widehat{F}(\sigma)$ is the inner product with χ_σ under that uniform measure. In particular, further smoothing decreases every influence.

Lemma A.2 (Uniform collision estimate)**.** *For every $0 < \kappa < 1/2$, there is $\mu_0(\kappa) > 0$. For each $m \ge 2$, there is a nonnegative function $r_c^{m,\kappa} : (0,\infty) \to [0,\infty)$, tending to zero as $t \downarrow 0$, with the following property, uniformly over all $R \ge 1$. Let $t > 0$, and let $F, G : [m]^R \to [0,1]$ have the same mean $0 < \mu \le \mu_0(\kappa)$ and satisfy*

$$\mathrm{Inf}_\ell(T_{1-\kappa}F) \le t, \qquad \mathrm{Inf}_\ell(T_{1-\kappa}G) \le t \quad (\ell \in [R]).$$

For every coupling (Z, Z') whose two marginals are uniform on $[m]^R$,

$$\mathbb{E}\big[(T_{1-2\kappa}F)(Z)(T_{1-2\kappa}G)(Z')\big] \le \mu^{1+\kappa/2} + r_c^{m,\kappa}(t). \tag{A.9}$$

The function $r_c^{m,\kappa}$ is independent of R, μ, F, G and the coupling.

Proof. This is the dimension-uniform form of [17, Lemmas 3.4 and 3.5] needed here. For $H = T_{1-\kappa}F$, the Fourier formula gives

$$\|T_{1-2\kappa}F\|_2^2 \le \mathrm{Stab}_{1-\kappa}(H),$$

where $\mathrm{Stab}_\theta(H) = \sum_\sigma \theta^{|\sigma|}\widehat{H}(\sigma)^2$. Indeed, $(1-\kappa)^3 - (1-2\kappa)^2 = \kappa(1-\kappa-\kappa^2) > 0$. Each full coordinate influence bound for H implies the corresponding low-degree influence bound in Theorem 4.4 of Mossel, O'Donnell, and Oleszkiewicz [32]. Apply that theorem with product dimension R, one-coordinate atom bound $1/m$, and correlation $1-\kappa$. For all sufficiently small t, it gives

$$\mathrm{Stab}_{1-\kappa}(H) \le \Gamma_{1-\kappa}(\mu) + C\frac{\log(2m)}{\kappa}\frac{\log\log(1/t)}{\log(1/t)}.$$

The cutoff and this error are independent of R and μ. Here $\Gamma_\theta(\mu)$ is the Gaussian halfspace noise stability of a set of Gaussian measure μ.

For fixed κ, Theorem B.5 of the same source gives $\Gamma_{1-\kappa}(\mu) = \mu^{2/(2-\kappa)+o(1)}$ as $\mu \downarrow 0$. Since

$$\frac{2}{2-\kappa} - \left(1 + \frac{\kappa}{2}\right) = \frac{\kappa^2}{2(2-\kappa)} > 0,$$

there is a cutoff $\mu_0(\kappa) > 0$, independent of m, R, below which $\Gamma_{1-\kappa}(\mu) \le \mu^{1+\kappa/2}$. This proves the corresponding squared L^2 bound for both F and G, with an error $r_c^{m,\kappa}(t) \to 0$ independent of the stated variables. Choose a cutoff $t_0 < e^{-e}$ within the range of the displayed bound, use its nonnegative error for $0 < t < t_0$, and set $r_c^{m,\kappa}(t) = 1$ for $t \ge t_0$. The latter range follows trivially from the product being at most one. Cauchy–Schwarz under any coupling with the specified marginals proves (A.9). □

Write $\mathcal{S}_q = \{p \in [0,1]^q : \sum_j p_j = 1\}$ and define the multilinear extension of $\overline{P_{\mathrm{id}}}$ on $[q]^k$ by

$$P_{\mathrm{id}}^{\mathrm{ml}}(p_1, \dots, p_k) = \sum_{r \in [q]^k} \overline{P_{\mathrm{id}}}(r) \prod_{i=1}^{k} p_i^{r_i}, \qquad p_i \in \mathcal{S}_q. \tag{A.10}$$

Lemma A.3 (A simplex-valued local estimate)**.** *Let J, μ_c, α be the data of Lemma A.1, and fix $0 < \lambda < 1$. For an integer $R \ge 1$, choose c with probability ω_c. Conditional on this one choice of c, choose the R coordinate tuples independently from μ_c, producing strings $Z_1, \dots, Z_k \in [m]^R$. Then obtain $\widetilde{Z}_i$ by independently resampling every position and coordinate uniformly with probability λ. Let $F = (F^1, \dots, F^q) : [m]^R \to \mathcal{S}_q$ be one function used for every occurrence c and every position. If $0 < t < 1/(kq)$ and*

$$\mathrm{Inf}_\ell(T_{1-\lambda} F^j) \le t \quad (\ell \in [R],\ j \in [q]),$$

then

$$\mathbb{E}_{c,Z,\widetilde{Z}} P_{\mathrm{id}}^{\mathrm{ml}}(F(\widetilde{Z}_1), \dots, F(\widetilde{Z}_k)) \le \mathrm{opt}_q(J) + \epsilon_{\mathrm{inv}}(kqt), \tag{A.11}$$

where $\epsilon_{\mathrm{inv}}(u) \to 0$ as $u \downarrow 0$, independently of R. The function ϵ_{inv} depends only on the fixed local data, k, q, λ.

Proof. Apply [17, Lemma 10.5] to the instance J, with local laws μ_c, noise parameter $\varepsilon = \lambda$, and payoff $P_{\mathrm{id}}^{\mathrm{ml}}$. The source's simplex N_q is our $\mathcal{S}_q$. Its test chooses one occurrence c, samples R independent tuples from μ_c, and independently noises each position, using the same function F throughout. This is exactly the experiment in the statement. Lemma A.1 supplies the uniform marginals, positive atom bound and common feasible Gram data required there.

Put $H = T_{1-\lambda} F$. Vector influence is the sum of the component influences [21, Definition 2.6], so

$$\mathrm{Inf}_\ell(H) = \sum_{j=1}^{q} \mathrm{Inf}_\ell(H^j) \le qt.$$

Thus the vector influence hypothesis of the cited lemma holds. Its error is uniform in R; we record the quantitative bound from its proof. For each c, the uniform one-position marginals and independent coordinate tuples imply that the influence of $H^j(Z_i)$, computed on $\mu_c^{\otimes R}$, equals that of H^j on $[m]^R$. The vector of all kq query components therefore has coordinate influences at most kqt.

In the cited proof, take Euclidean projection onto $\mathcal{S}_q$, which is nonexpansive by [21, Lemma 3.8]. Telescoping the k arguments of the multilinear payoff bounds its change on the simplex by $\sum_i \|x_i -$

$y_i\|_1$; hence its composition with these projections is $\sqrt{kq}$-Lipschitz. The invariance bound in [17, Theorem 10.1 and the proof of Lemma 10.5] therefore permits

$$\epsilon_{\mathrm{inv}}(u) = C_{kq}\sqrt{kq}\, u^{\lambda/(18\log(1/\alpha))}.$$

Here C_{kq} is independent of R, and all other parameters are fixed. The argument of Lemma 10.5, with this error in place of the unspecified error in its statement, gives (A.11); the error tends to zero as required. □

Proof of Proposition 9.3. We first derive estimates that hold for every R, then choose their numerical parameters independently of R. Fix for now $0 < \kappa < 1/2$, q with $1/q \le \mu_0(\kappa)$, and local data from Lemma A.1, and write

$$v_{\mathrm{loc}} = \sum_c \omega_c \mathbb{E}_{X\sim\mu_c} \overline{P_{\mathrm{id}}}(X).$$

Use the local experiment of Lemma A.3 with the actual resampling probability $\lambda = 2\kappa$. Also fix auxiliary constants $0 < t < 1/(kq)$ and $\chi > 0$.

For an input graph Φ, let $\pi_{a\to b}$ denote the permutation on edge ab. Its action on strings is fixed by

$$(\pi_{a\to b} z)_{\pi_{a\to b}(\ell)} = z_\ell \qquad (\ell \in [R]).$$

The output variable set will be $\mathcal{Y} = B \times [m]^R$. Consider the following independent neighbor experiment: choose a uniformly from A, choose $b_1, \dots, b_k$ independently and uniformly from $N(a)$, and independently perform the local experiment, obtaining $\widetilde{Z}_1, \dots, \widetilde{Z}_k$. Its raw output positions are

$$w_i = (b_i, \pi_{a\to b_i}\widetilde{Z}_i) \in \mathcal{Y}. \tag{A.12}$$

For any rank map $f : \mathcal{Y} \to \mathbb{Z}$, define

$$V(f) = \mathbb{E}\,\overline{P_{\mathrm{id}}}(f(w_1), \dots, f(w_k)), \qquad V_a(f) = \mathbb{E}[\overline{P_{\mathrm{id}}}(f(w_1), \dots, f(w_k)) \mid a].$$

The bounded payoff (A.4) is used even if the experiment repeats an output ID. We will obtain a lower bound for one coarse rank map and an upper bound for every injective order, then transfer both bounds to distinct-variable constraints.

Suppose a labeling L strongly satisfies at least $1 - \zeta$ of the left vertices. Define the coarse rank map $f(b, z) = z_{L(b)}$. At every strongly satisfied a and every neighbor b,

$$f(b, \pi_{a\to b} z) = z_{\pi_{a\to b}^{-1}(L(b))} = z_{L(a)}.$$

Thus all k positions read the same label coordinate, for every choice of neighbors. With probability $(1 - 2\kappa)^k$ none of these k symbols is resampled; conditional on that event their tuple has law μ_c. Nonnegativity of the payoff gives

$$V(f) \ge (1 - \zeta)(1 - 2\kappa)^k v_{\mathrm{loc}}. \tag{A.13}$$

For soundness fix an injective order $O : \mathcal{Y} \to [|\mathcal{Y}|]$. At a fixed a, its local set $N(a) \times [m]^R$ has exactly $M = hm^R$ distinct IDs. Sort this set by O and divide the sorted list into q consecutive blocks $C_{a,1}, \dots, C_{a,q}$ of size M/q. These sizes are integers because $q \mid h$. Each block is the intersection of the local set with an interval of global ranks. Define

$$F_a^j(z) = \frac{1}{h} \sum_{b\in N(a)} \mathbf{1}_{\{(b,\pi_{a\to b} z)\in C_{a,j}\}}, \qquad F_a = (F_a^1, \dots, F_a^q). \tag{A.14}$$

For every z, $F_a(z) \in \mathcal{S}_q$. Each edge permutation is a bijection on strings, so

$$\mathbb{E}_z F_a^j(z) = \frac{|C_{a,j}|}{hm^R} = \frac{1}{q}. \tag{A.15}$$

For any interval I of integer ranks put

$$F_a^I(z) = \frac{1}{h} \sum_{b \in N(a)} \mathbf{1}_{\{O(b,\pi_{a\to b}z) \in I\}}.$$

Call *a pseudorandom at threshold* t if $\mathrm{Inf}_\ell(T_{1-\kappa}F_a^I) \le t$ for every I and every $\ell \in [R]$. At such an a, this bound holds for each block function F_a^j.

Let $j_a(w)$ be the block index of a local ID w. Conditional on the local query strings, independence of the neighbor choices gives the exact identity

$$\mathbb{E}_{b_1,\dots,b_k} \overline{P_{\mathrm{id}}}(j_a(w_1),\dots,j_a(w_k)) = P_{\mathrm{id}}^{\mathrm{ml}}(F_a(\widetilde{Z}_1),\dots,F_a(\widetilde{Z}_k)). \tag{A.16}$$

The identity includes outcomes with equal neighbors. Further smoothing decreases the block influences, so $\mathrm{Inf}_\ell(T_{1-2\kappa}F_a^j) \le t$. Lemma A.3, applied to the single function F_a, bounds the expected coarse payoff in (A.16) by $\mathrm{opt}_q(J) + \epsilon_{\mathrm{inv}}(kqt)$.

When the k coarse ranks are distinct, their induced permutation is the one induced by the fine ranks $O(w_i)$. Otherwise the two payoffs differ by at most one. This also covers a repeated output ID, which necessarily produces a coarse tie. For two positions $i \ne i'$, independence of their noise and neighbor choices gives their coarse tie probability as

$$\sum_{j=1}^{q} \mathbb{E}_c \mathbb{E}_{Z\sim\mu_c^{\otimes R}}\left[(T_{1-2\kappa}F_a^j)(Z_i)(T_{1-2\kappa}F_a^j)(Z_{i'})\right].$$

Conditional on c, both strings have uniform product marginals. Their correlation is permitted by Lemma A.2. Use that lemma with the fixed mean $1/q$ from (A.15). The last display is at most $q^{-\kappa/2} + qr_c^{m,\kappa}(t)$. A union bound over pairs therefore gives, at every pseudorandom a,

$$V_a(O) \le S_{\mathrm{loc}} := \mathrm{opt}_q(J) + \binom{k}{2}(q^{-\kappa/2} + qr_c^{m,\kappa}(t)) + \epsilon_{\mathrm{inv}}(kqt). \tag{A.17}$$

We next turn an excess over this bound into a labeling of Φ. For each $b \in B$, restrict O to $\{b\} \times [m]^R$ and compress its ranks to obtain a strict order o_b of $[m]^R$. Let

$$\mathcal{L}_b = \left\{\ell \in [R] : \mathrm{Inf}_\ell(T_{1-\kappa}\mathbf{1}_{\{o_b(z)\in I\}}) > t/2 \text{ for some rank interval } I\right\}.$$

The few-influential-coordinates lemma [17, Definition 4.2 and Lemma 4.3] gives

$$|\mathcal{L}_b| \le \frac{400}{\kappa(t/2)^3} = \frac{3200}{\kappa t^3}. \tag{A.18}$$

This bound is uniform in R.

If a is not pseudorandom, choose one witnessing interval I_a and coordinate ℓ_a with $\mathrm{Inf}_{\ell_a}(T_{1-\kappa}F_a^{I_a}) > t$. For any coordinate permutation π, scalar function f, and $0 \le \theta \le 1$, independent coordinate resampling and uniform measure give

$$T_\theta(f\circ\pi) = (T_\theta f)\circ\pi, \qquad \mathrm{Inf}_\ell(f\circ\pi) = \mathrm{Inf}_{\pi(\ell)}(f).$$

The direction of the index here uses the declared action $(\pi z)_{\pi(\ell)} = z_\ell$. Also $\mathrm{Inf}_\ell(f) = \|f - \mathbb{E}_{z_\ell} f\|_2^2$, so Jensen's inequality for this squared norm applies to the neighbor average. Linearity of noise then gives

$$\mathrm{Inf}_{\ell_a}(T_{1-\kappa} F_a^{I_a}) \le \frac{1}{h} \sum_{b \in N(a)} \mathrm{Inf}_{\pi_{a\to b}(\ell_a)} \left(T_{1-\kappa} \mathbf{1}_{\{O(b,z) \in I_a\}} \right). \tag{A.19}$$

Each term is at most one. An average greater than t forces more than a $t/2$ fraction of the terms to exceed $t/2$. The restriction of a global-rank interval to one right code is an interval of its relative order, so these neighbors satisfy $\pi_{a\to b}(\ell_a) \in \mathcal{L}_b$.

Give each such a label ℓ_a, and label every other left vertex arbitrarily. Independently at each right vertex with $\mathcal{L}_b \neq \varnothing$, choose one uniform label from $\mathcal{L}_b$; use any label when the set is empty. By (A.18), the expected satisfied fraction among the edges at each nonpseudorandom a is greater than

$$\frac{t}{2} \frac{\kappa t^3}{3200} = \frac{\kappa t^4}{6400}.$$

One random right labeling is used for all left vertices. Linearity of expectation and the common left degree identify the resulting average with the ordinary uniform edge fraction.

If $V(O) > S_{\mathrm{loc}} + \chi$, more than a χ fraction of left vertices are nonpseudorandom: if that fraction is g, the bound $0 \le V_a(O) \le 1$ and (A.17) give $V(O) \le (1-g)S_{\mathrm{loc}} + g \le S_{\mathrm{loc}} + g$. The preceding decoder would then give a labeling of edge value greater than $\chi\kappa t^4/6400$. Consequently

$$\mathrm{val}(\Phi) \le \nu < \frac{\chi\kappa t^4}{6400} \quad \Longrightarrow \quad V(O) \le S_{\mathrm{loc}} + \chi \quad \text{for every injective } O. \tag{A.20}$$

It remains to turn the independent experiment into strict ordering constraints. Let D be the event that $b_1, \dots, b_k$ are distinct. For $h \ge k$,

$$p_D := \Pr(D) = \frac{(h)_k}{h^k}, \qquad \delta_h := 1 - p_D \le \frac{\binom{k}{2}}{h}. \tag{A.21}$$

The probability is the same at every a, and D is independent of the local experiment. Conditional on D, a remains uniform and its ordered distinct neighbor tuple is uniform among its $(h)_k$ possibilities. All output IDs are then distinct because their first coordinates b_i differ. Let I_0 denote this finite weighted distribution of strict P_{id} constraints, and write $\mathrm{OPT}_{P_{\mathrm{id}}}(I_0)$ for its maximum strict payoff.

Let $V_D(f)$ denote the expected tie-extended payoff after this conditioning, and use $V_{\neg D}(f)$ for conditioning on its complement. For any rank map f, coarse or injective,

$$V(f) = p_D V_D(f) + (1 - p_D) V_{\neg D}(f), \qquad |V_D(f) - V(f)| \le \delta_h, \tag{A.22}$$

because all payoffs lie in $[0,1]$. Apply this first to the coarse map in (A.13). Within each finite fiber of that map, choose one uniform random permutation of the distinct IDs, independently between fibers, and concatenate the fibers in rank order. On every conditioned tuple, restriction of these permutations to its IDs is uniform within each tied rank class. Thus the expected strict P_{id}-payoff of this one random global order equals $V_D(f)$. Some global order attains at least that value. For soundness, an injective O has distinct ranks on every conditioned tuple, so its tie-extended payoff there is its strict P_{id}-payoff. Equations (A.13), (A.20), and (A.22) therefore give the strict bounds

$$\begin{aligned} \text{YES}: &\quad \mathrm{OPT}_{P_{\mathrm{id}}}(I_0) \ge (1-\zeta)(1-2\kappa)^k v_{\mathrm{loc}} - \delta_h, \\ \text{NO}: &\quad \mathrm{OPT}_{P_{\mathrm{id}}}(I_0) \le S_{\mathrm{loc}} + \chi + \delta_h \quad \text{if } \nu < \chi\kappa t^4/6400. \end{aligned} \tag{A.23}$$

These bounds are for the conditioned P_{id} tuples. Applying (A.5) makes every occurrence use the original predicate Π with exactly the same values. Call this emitted instance I.

We now make the choices for the prescribed δ. Choose a positive rational η with $10\eta < \delta$, and a rational $0 < \kappa < 1/2$ with $2\kappa k < \eta$. Put $N_2 = \binom{k}{2}$. Choose an integer q so large that

$$\frac{1}{q} \le \mu_0(\kappa), \qquad N_2 q^{-\kappa/2} < \eta.$$

Choose the data J, m, α from Lemma A.1 at error η. The two analytic error functions now have fixed parameters and are uniform in R. Choose a positive rational $t < 1/(kq)$ so small that

$$N_2 q r_c^{m,\kappa}(t) < \eta, \qquad \epsilon_{\mathrm{inv}}(kqt) < \eta.$$

Set $\chi = \eta$. Choose positive rationals $\zeta, \nu < 1$ with

$$\zeta < \eta, \qquad \nu < \frac{\eta\kappa t^4}{6400},$$

and choose an integer $H \ge \max\{k, q\}$ with $N_2/H < \eta$. These choices depend only on k, Π, δ. In particular, they precede both the choice of h in Lemma 9.1 and the alphabet supplied by Theorem 1.1.

For every $h \ge H$, $\delta_h < \eta$. The local gap (A.6), Bernoulli's inequality, and (A.23) give

$$\text{YES}: \quad \mathrm{OPT}_\Pi(I) \ge 1 - \zeta - 2\kappa k - \eta - \delta_h > 1 - 4\eta > 1 - \delta,$$

whereas (A.17) gives

$$\text{NO}: \quad \mathrm{OPT}_\Pi(I) \le S_{\mathrm{loc}} + \chi + \delta_h < \rho_\Pi + 6\eta < \rho_\Pi + \delta.$$

This proves the numerical claims simultaneously for every R.

Finally we specify the encoded transformation. Once R is fixed, the choice of c, the product $\mu_c^{\otimes R}$, and the independent rational 2κ-noise form one fixed finite rational law θ_R on local outcomes ξ. These are the actual local outcomes above. Renumber B by $[|B|]$ and explicitly list $\mathcal{Y} = B \times [m]^R$. For each $a \in A$, each ordered distinct neighbor tuple, and each $\xi \in \operatorname{supp} \theta_R$, list the tuple in (A.12), permuted by (A.5), with weight

$$\frac{\theta_R(\xi)}{|A|(h)_k}. \tag{A.24}$$

The weights sum to one and are nonnegative rationals. Every listed tuple has distinct IDs; the same tuple may occur in several entries. There are $|B|m^R$ variables and at most $|A|(h)_k|\operatorname{supp}\theta_R|$ entries. For fixed k, m, R, θ_R, these counts are polynomial in the explicit graph size, even when h varies. A fixed common denominator for θ_R shows that each weight has $O(\log|A| + k\log h)$ bits plus a fixed constant. The IDs have $O(\log|B|)$ bits plus a fixed constant. Exact enumeration of these entries is therefore deterministic and polynomial for fixed R, as asserted. □

The same occurrence-list encoding also justifies the deterministic random-order guarantee in Section 9. Build an order by exposing one next variable at a time. If exactly s variables of a constraint have been exposed, its conditional success probability is the number of permutations in Π whose first s positions are the roles of those variables in the exposed order, divided by $(k-s)!$. It can be computed by inspecting the fixed set S_k. For each choice of the next variable, sum these probabilities against the listed rational weights and choose a next variable whose conditional expectation is at least the average over the choices. This preserves the initial expectation $|\Pi|/k!$. All comparisons are exact and polynomial in the input bit length: the product of the weight denominators has bit length at most their total encoded length, and the factorial denominators have constant size for fixed k. The variable list is explicit, so the number of prefix steps and candidate choices is polynomial as well.

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